



P802.1ASds/D1.1

March 23, 2025

(Amendment to IEEE Std 802.1AS™-2025)

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Draft Standard for Local and metropolitan area networks— Timing and Synchronization for Time-Sensitive Applications

Amendment: Support for the IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex

12 Sponsor

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15 IEEE Computer Society

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4 Information on participation in this project, and in the IEEE 802.1 Working Group can be found [here](#).

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23 802.1 Working Group.

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33 referred to as the 5 Criteria or 5C's) is reviewed on a regular basis in order to ensure their continued validity.

34 A vote of "Approve" on this draft is also an affirmation by the balloter that the PAR is still valid.

1 **Project Authorization Request, Scope, Purpose, and Criteria for Standards** 2 **Development (CSD)**

3 The complete amendment PAR, as approved by IEEE NesCom 23 February 2022, can be found at:

4 <https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/9522>

5 The ‘Scope of the Proposed changes’ and the ‘Need for the Project’ specify the changes to be made by this
6 amendment (see below).

7 **Scope of the Proposed changes:**

8 This amendment specifies protocols, procedures, and managed objects that support IEEE Std 802.3 Clause 4
9 Media Access Control (MAC) operating in half-duplex while retaining existing functionality and backward
10 compatibility, and remaining a profile of IEEE Std 1588™-2019.

11 This amendment addresses errors and omissions in the description of existing functionality.

12 **Need for the Project:**

13 Support is needed in applications such as automotive in-vehicle networks and industrial automation networks
14 for the IEEE Std 802.3 Clause 4 MAC operating in half-duplex, including those using links with the
15 10BASE-T1S PHY in either point-to-point or multidrop half-duplex mode recently introduced by IEEE Std
16 802.3cg-2019.

17 **Criteria for Standards Development:**

18 The complete Criteria for Standards Development (CSD) can be found at:

19 <https://mentor.ieee.org/802-ec/dcn/21/ec-21-0308-00-ACSD-p802-1asds.pdf>

20

P802.1ASds/D1.1
March 23, 2025

(Amendment to IEEE Std 802.1AS™-202x)

Draft IEEE Standard for Local and metropolitan area networks—

Timing and Synchronization for Time-Sensitive Applications

Amendment: Support for the IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex

Prepared by the
Time-Sensitive Networking (TSN) Task Group of IEEE 802.1

Sponsor
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1 **Abstract:** This amendment to IEEE Std 802.1AS™-2020 specifies protocols, procedures, and
2 managed objects that support IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in
3 half-duplex while retaining existing functionality and backward compatibility, and remaining a profile
4 of IEEE Std 1588™-2019.

5 This amendment addresses errors and omissions in the description of existing functionality.

6 **Keywords:** best timeTransmitter, frequency offset, Grandmaster Clock, Grandmaster PTP
7 Instance, PTP End Instance, PTP Relay Instance, IEEE Std 802.1AS™, phase offset,
8 synchronization, syntonization, time-aware system

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5 **Jessy Rouyer, *Vice Chair***
6 **János Farkas, *TSN Task Group Chair***
7 **Silvana Rodrigues, *Editor IEEE Std 802.1AS***
8 **Silvana Rodrigues, *Editor P802.1ASds***
9

10 The following members of the individual balloting committee voted on this standard. Balloters may have
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13

1

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8

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10

1 Introduction

This introduction is not part of IEEE Std 802.1ASdsTM-20xx, IEEE Standard for Local and metropolitan area networks—Timing and Synchronization for Time-Sensitive Applications—Amendment: Support for the IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex

2 The first edition of IEEE Std 802.1AS was published in 2011. A first corrigendum, IEEE Std
3 802.1ASTM-2011/Cor1-2013, provided technical and editorial corrections. A second corrigendum, IEEE Std
4 802.1ASTM-2011/Cor2-2015 provided additional technical and editorial corrections.

5 The second edition, IEEE Std 802.1AS-2020, added support for multiple gPTP domains, Common Mean
6 Link Delay Service, external port configuration, and Fine Timing Measurement for 802.11 transport.
7 Backward compatibility with IEEE Std 802.1AS-2011 was maintained. A corrigendum, IEEE Std
8 802.1ASTM-2020/Cor1-2021, provides technical and editorial corrections.

9 The third edition, IEEE Std 802.1AS-202x is a roll-up of IEEE Std 802.1AS-2020 with the corrigendum
10 IEEE Std 802.1AS-2020/Cor1, and its amendments: IEEE Std 802.1ASdr, IEEE Std 802.1ASdn, and IEEE
11 Std 802.1ASdm.

12 This amendment to IEEE Std 802.1AS-202x specifies protocols, procedures, and managed objects that
13 support IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex while retaining
14 existing functionality and backward compatibility, and remaining a profile of IEEE Std 1588TM-2019.

15 This amendment addresses errors and omissions in the description of existing functionality

16 <<Editor's note: P802.1ASds is an amendment to 802.1AS-2025 that is the outcome of
17 P802.1AS-2020-REV.>>

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1

2 **Draft IEEE Standard for** 3 **Local and metropolitan area networks—**

4 **Timing and Synchronization for Time-** 5 **Sensitive Applications**

6 **Amendment: Support for the IEEE Std 802.3** 7 **Clause 4 Media Access Control (MAC) operating in** 8 **half-duplex**

9 [This amendment is based on IEEE Std 802.1AS™-20xx (IEEE Std 802.1AS™-2020 Revision).

10 NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into
11 the existing base standard and its amendments to form the comprehensive standard.

12 The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace.
13 ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change
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18 one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the
19 changes will be incorporated into the base standard.¹

20

¹Notes in text, tables, and figures are given for information only, and do not contain requirements needed to implement the standard.

3. Definitions

Insert the following definitions in Clause 3, and renumber the definitions as appropriate:

3.17 Half-duplex Ethernet: An Ethernet whose physical ports use the IEEE Std 802.3TM Clause 4 MAC in half-duplex mode, independent of the physical medium access method and independent of the physical topology of the medium connected (which is typically a shared medium).

1 **4. Acronyms and abbreviations**

2 *Insert the following acronym in clause 4 as follows:*

3 HDE Half-duplex Ethernet

4

5. Conformance

5.5 .MAC-specific timing and synchronization methods for full-duplex IEEE 802.3 links

Change the first sentence of 5.5 as follows:

An implementation of a time-aware system with full-duplex IEEE 802.3 media access control (MAC) services to physical ports shall:

Insert 5.9 as follows:

5.9 MAC-specific timing and synchronization methods for Half-duplex Ethernet (HDE) links

An implementation of a time-aware system with half-duplex IEEE 802.3 media access control (MAC) services to physical ports shall:

- a) Support half-duplex operation, as specified in Clause 4 of IEEE Std 802.3.
- b) Support the requirements as specified in Clause 19.

14

1 7. Time-synchronization model for a packet network

2 7.2 Architecture of a time-aware network

3 7.2.1 General

4 *Add item g) after item f) in the lettered list in 7.2.1 as follows:*

- 5 g) IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex (Clause 19)

6 7.3 Time synchronization

7 7.3.2 Delay measurement

8 *Add item e) after item d) in the lettered list in 7.3.2 as follows:*

- 9 e) HDE uses the same mechanism as full-duplex Ethernet

10 7.4 PTP Instance architecture

11 *Change the last paragraph of 7.4 as follows:*

12 For EPON, timing information is communicated using a “slow protocol” as defined in Clause 13. CSNs and
13 HDE use the same communication system used by full-duplex Ethernet, as defined in Clause 16 and Clause
14 19, respectively.

15 7.5 Differences between gPTP (IEEE Std 802.1AS) and PTP (IEEE Std 1588-2019)

16 *Change item e) and item f) in 7.5 as follows:*

- 17 e) For full-duplex Ethernet links and HDE, gPTP requires the use of the peer-to-peer delay mechanism,
18 while IEEE Std 1588-2019 also allows the use of end-to-end delay measurement for full-duplex
19 Ethernet links.
- 20 f) For full-duplex Ethernet links and HDE, gPTP requires the use of two-step processing (use of
21 Follow_Up and Pdelay_Resp_Follow_Up messages to communicate timestamps), with an optional
22 one-step processing mode for full-duplex Ethernet that embeds timestamps in the Sync “on the fly”
23 as they are being transmitted (gPTP does not specify one-step processing for peer delay messages).
24 IEEE Std 1588-2019 allows either two-step or one-step processing to be required (for both Sync and
25 peer delay messages) depending on a specific profile.

1 8. IEEE 802.1AS concepts and terminology

2 8.5 Ports

3 8.5.1 General

4 *Change 8.5.1 as follows:*

5 The PTP Instances in a gPTP domain interface with the network media via physical ports. gPTP defines a
6 ~~logical port, i.e.,~~ a PTP Port, in such a way that communication between PTP Instances is point-to-point or,
7 point-to-multipoint. The choice of point-to-point (e.g., see Clause 16) or point-to-multipoint (e.g., see
8 Clause 19) communication is specified in each media dependent specification in this standard. even over
9 ~~physical ports that are attached to shared media. One logical~~ A PTP port, consists ing of one PortSync entity
10 and one media-dependent (MD) entity, ~~is instantiated for each PTP Instance with which the PTP Instance~~
11 ~~communicates. For shared media, multiple logical ports can be associated with a single physical port.~~
12 ~~Unless otherwise qualified, each instance of the term port refers to a logical port.~~

1 11. Media-dependent layer specification for full-duplex point-to-point links

2 11.1 Overview

3 11.1.1 General

4 *Add a NOTE at the end of 11.1.1 as follows:*

5 NOTE—PTP links using the IEEE Std 802.3 Clause 4 MAC operating in half-duplex mode are specified in Clause
6 19.Determination of asCapable and asCapableAcrossDomains.

7 11.2 State machines for MD entity specific to full-duplex point-to-point links

8 11.2.1 General

9 *Replace Figure 11-5 with the following:*

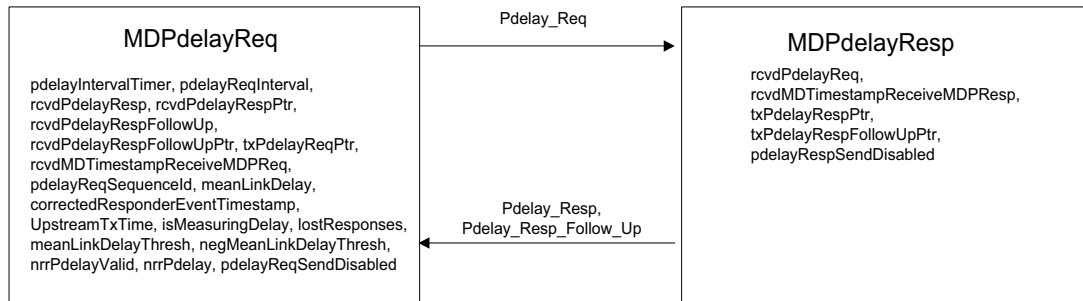


Figure 11-5—Peer-to-peer delay mechanism state machines—
overview and interrelationships

10 11.2.2 Determination of asCapable and asCapableAcrossDomains

11 *Change 11.2.2 as follows:*

12 There is one instance of the global variable asCapable (see 10.2.5.1) per PTP Port, per domain. For full-
13 duplex point-to-point links, there is one instance of the global variable asCapableAcrossDomains (see
14 11.2.13.12), per port, that is common across, and accessible by, all the domains. For HDE, there is one
15 instance of this variable per PTP Instance (i.e., per domain) (see 11.2.13.12).

16 The ~~per-PTP Port~~-global variable asCapable (see 10.2.5.1) indicates whether the IEEE 802.1AS protocol is
17 operating, in this domain, on the PTP Link attached to this PTP Port, and can provide the required time-
18 synchronization performance ~~described in B.3~~. asCapable is used by the PortSync entity, which is media-
19 independent; however, the determination of asCapable is media-dependent.

20 The per-port global variable asCapableAcrossDomains is set by the MDPdelayReq state machine
21 (see 11.2.19 and Figure 11-9). For a port attached to a full-duplex point-to-point PTP Link or to an HDE
22 link, asCapableAcrossDomains shall be set to TRUE if and only if either:

- 1 a) ~~It~~ is determined, via the transport-specific peer-to-peer delay mechanism or CMLDS, that the
 2 following conditions hold for the port:
- 3 1) ~~⊕~~The port is exchanging peer delay messages with its neighbor,
 - 4 2) ~~⊕~~The measured delay does not exceed meanLinkDelayThresh__or is below
 5 negMeanLinkDelayThresh,
 - 6 3) ~~⊕~~The port does not receive multiple Pdelay_Resp or Pdelay_Resp_Follow_Up messages in
 7 response to a single Pdelay_Req message, and
 - 8 4) ~~⊕~~The port does not receive a response from itself or another PTP Port of the same PTP
 9 Instance.

10 or:

- 11 b) pdelayReqSendDisabled is set to TRUE

12 NOTE 1—If a PTP Instance implements only domain 0 and the MDPdelayReq and MDPdelayResp state machines are
 13 invoked on domain 0 (see 11.2.19), asCapableAcrossDomains is still set by the MDPdelayReq state machine.

14 The default value of meanLinkDelayThresh shall be set as specified in Table 11-1.

Table 11-1—Value of meanLinkDelayThresh for various links

Link	Value of meanLinkDelayThresh (ns) (see NOTE)
100BASE-TX, 1000BASE-T	800 ₁₀
100BASE-FX, 1000BASE-X	0x7FFFFFFF
NOTE—The actual propagation delay for 100BASE-TX and 1000BASE-T links is expected to be smaller than the above respective threshold. If the measured mean propagation delay (i.e., meanLinkDelay; see 10.2.5.8) exceeds this threshold, it is assumed that this is due to the presence of equipment that does not implement gPTP. For 100BASE-FX and 1000BASE-X links, the actual propagation delay can be on the order of, or larger than, the delay produced by equipment that does not implement gPTP; therefore, such equipment cannot be detected by comparing measured propagation delay with a threshold. In this case, meanLinkDelayThresh is set to the largest possible positive value (i.e., in binary notation a zero followed by all 1s).	

15 The per-PTP Port, per-domain global variable asCapable shall be set to TRUE if and only if the following
 16 conditions hold:

- 17 c) ~~⊕~~The value of asCapableAcrossDomains is TRUE, and
- 18 d) ~~⊕~~One of the following conditions holds:
 - 19 1) The value of neighborGptpCapable for this PTP Port is TRUE, or
 - 20 2) The value of domainNumber is zero, and the value of sdoId for peer delay messages received
 21 on this PTP Port is 0x100.

22 NOTE 2—Condition ~~⊕~~d 2) ensures backward compatibility with the 2011 edition of this standard. A PTP Instance
 23 compliant with the current edition of this standard that is attached, via a full-duplex point-to-point PTP Link, to a PTP
 24 Instance compliant with the 2011 edition of this standard will not receive Signaling messages that contain the gPTP
 25 capable TLV and will not set neighborGptpCapable to TRUE. However, condition ~~⊕~~d 2) ensures that asCapable for
 26 this PTP Port and domain (i.e., domain 0) will still be set to TRUE if condition c) holds because the peer delay messages
 27 received from the time-aware system compliant with the 2011 edition of this standard will have sdoId set to 0x100.

1 11.2.13 MD entity global variables**2 Change 11.2.13.12 as follows:**

3 **11.2.13.12 asCapableAcrossDomains:** A Boolean that is TRUE if and only if either: 11.2.2 item a) or
 4 11.2.2 item b) is satisfied~~conditions a) through d) of 11.2.2 are satisfied. This Boolean~~
 5 asCapableAcrossDomains is set by the MDPdelayReq state machine and is used in determining asCapable
 6 for a port (see 11.2.2). There is one instance of this variable for all the domains (per port) for full-duplex
 7 point-to-point links.~~And~~ the variable is accessible by all the domains. There is one instance of this variable
 8 per PTP Instance (i.e., per domain) for HDE (see 19.2.2). When only one domain is active,
 9 asCapableAcrossDomains is equivalent to the variable asCapable (see 10.2.5.1).

10 11.2.19 MDPdelayReq state machine**11 11.2.19.2 State machine variables****12 Change 11.2.19.2.2 as follows:****13 11.2.19.2.2 rcvdPdelayResp:**

- 14 a) For full-duplex point-to-point mode: A Boolean variable that notifies the current state machine
 15 when a Pdelay_Resp message is received. This variable is reset by the current state machine.
- 16 b) For HDE mode: A Boolean variable that notifies the current state machine when a Pdelay_Resp
 17 message is received AND its requestingPortIdentity.clockIdentity is equal to the clockIdentity of the
 18 current PTP Instance. This variable is reset by the current state machine.

19 Change 11.2.19.2.4 as follows:**20 11.2.19.2.4 rcvdPdelayRespFollowUp:**

- 21 a) For full-duplex point-to-point mode: A Boolean variable that notifies the current state machine
 22 when a Pdelay_Resp_Follow_Up message is received. This variable is reset by the current state
 23 machine.
- 24 b) For HDE mode: A Boolean variable that notifies the current state machine when a
 25 Pdelay_Resp_Follow_Up message is received AND its requestingPortIdentity.clockIdentity is equal
 26 to the clockIdentity of the current PTP instance. This variable is reset by the current state machine.

27 Insert a new variable after 11.2.19.2.13 as follows:

28 **11.2.19.2.14 pdelayReqSendDisabled:** A boolean that is administratively set to TRUE if Pdelay_Req
 29 messages are not transmitted by this port. The default value for this variable shall be FALSE.

30 11.2.19.4 State diagram**31 Replace Figure 11-9 with the following:.**

32

33

34

35

36

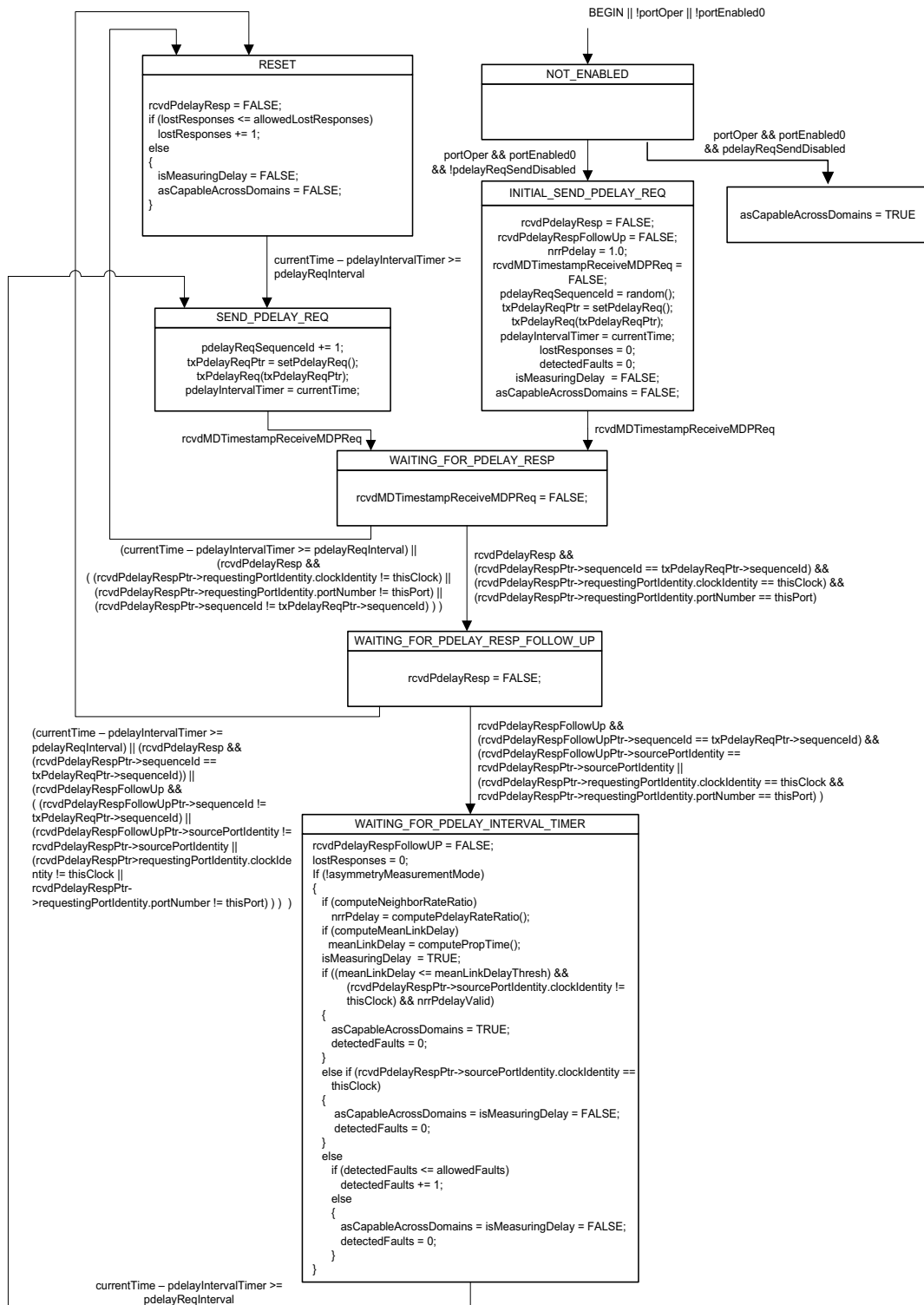


Figure 11-9—MDPdelayReq state machine

1 **Add a NOTE after Figure 11-9 as follows:**

2 NOTE—A change in the value of the variable pdelayReqSendDisabled takes effect only when portEnabled0 (see
3 11.2.19.2.12) is FALSE.

4 11.2.20 MDPdelayResp state machine

5 11.2.20.2 State machine variables

6 **Insert a new variable after 11.2.20.2.5 as follows:**

7 **11.2.20.2.6 pdelayRespSendDisabled:** A boolean that is administratively set to TRUE to indicate that
8 corresponding Pdelay_Resp and Pdelay_Resp_Follow_Up messages are not transmitted by this port in
9 response to a Pdelay_Req message. The default value for this variable shall be FALSE.

10 11.2.20.4 State diagram

11 **Replace Figure 11-10 with the following:**

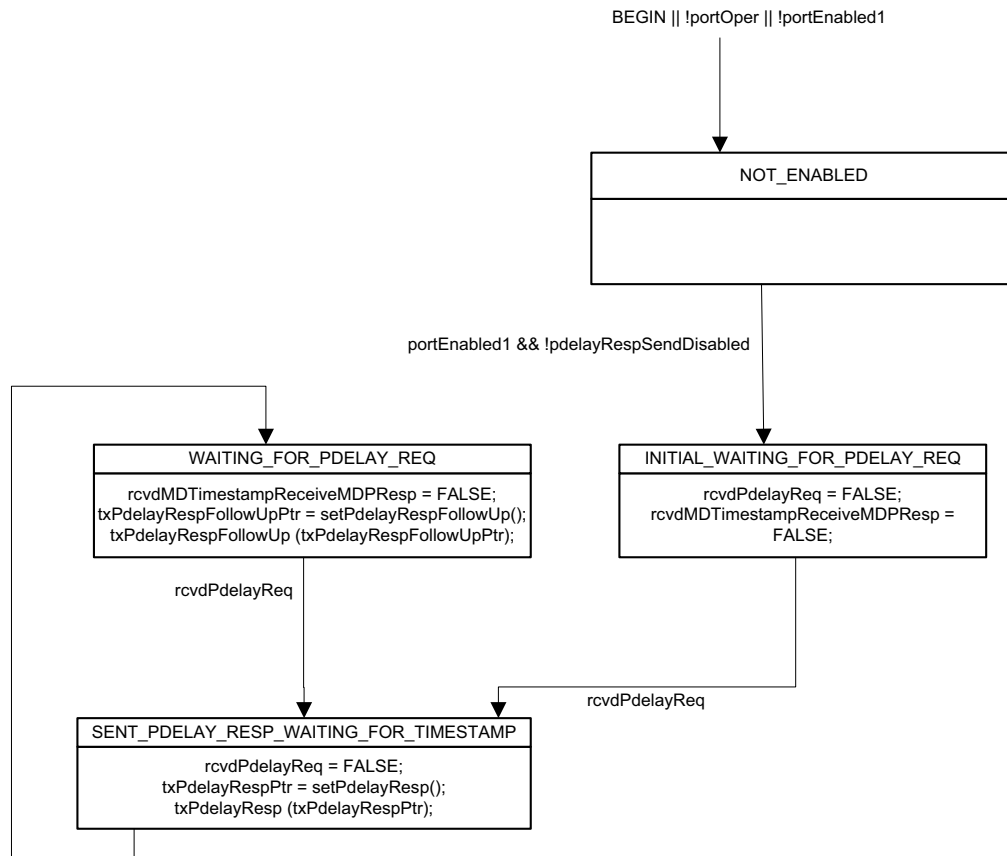


Figure 11-10—MDPdelayResp state machine

12 **Add a NOTE after Figure 11-10 as follows:**

¹ NOTE—A change in the value of the variable `pdelayRespSendDisabled` takes effect only when `portEnabled1` (see
² 11.2.20.2.5) is FALSE.

1 14. Timing and synchronization management

2 14.8 Port Parameter Data Set (portDS)

3 14.8.23 InitialLogPdelayReqInterval

4 *Change the first paragraph in 14.8.23 as follows:*

5 For full-duplex [and HDE](#) IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism
6 to measure path delay (see 16.4.3.2), the value is the logarithm to base 2 of the Pdelay_Req message
7 transmission interval used when:

8 14.8.24 currentLogPdelayReqInterval

9 *Change the first paragraph in 14.8.24 as follows:*

10 For full-duplex [and HDE](#) IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism
11 to measure path delay (see 16.4.3.2), the value is the logarithm to the base 2 of the current Pdelay_Req
12 message transmission interval (see 11.5.2.2).

13 14.8.32 currentComputeNeighborRateRatio

14 *Change the first paragraph in 14.8.32 as follows:*

15 For full-duplex [and HDE](#) IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism
16 to measure path delay (see 16.4.3.2), the value is the current value of computeNeighborRateRatio.

17 14.8.36 currentComputeMeanLinkDelay

18 *Change the first paragraph in 14.8.36 as follows:*

19 For full-duplex [and HDE](#) IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism
20 to measure path delay (see 16.4.3.2), the value is the current value of computeMeanLinkDelay.

21 14.8.53 pdelayTruncatedTimestampsArray

22 *Change the first paragraph in 14.8.53 as follows:*

23 For full-duplex [and HDE](#) IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism
24 to measure path delay (see 16.4.3.2), the values of the four elements of this array are as described in Table
25 14-9. For all other media, the values are zero. Array elements 0, 1, 2, and 3 correspond to the timestamps t1,
26 t2, t3, and t4, respectively, in Figure 11-1 and are expressed in units of 2–16 ns (i.e., the value of each array
27 element is equal to the remainder obtained upon dividing the respective timestamp, expressed in units of 2–
28 16 ns, by 248). At any given time, the timestamp values stored in the array are for the same, and most
29 recently completed, peer delay message exchange.

30 14.8.55 portDS table

31 *Insert the following items after the final item of Table 14-10:*

32

Table 14-12—portDS table

Name	Data type	Operations supported ^a	References
pdelayReqSendDisabled	Boolean	RW	14.8.60
pdelayRespSendDisabled	Boolean	RW	14.8.61

^a R = Read only access; RW = Read/write access.

Insert 14.8.60 and 14.8.61 as follows:

14.8.60 pdelayReqSendDisabled

The value is equal to the value of the per-PTP Port variable pdelayReqSendDisabled (see 11.2.19.2.14). If its value is TRUE, Pdelay_Req messages are not transmitted by the PTP Port. The default value for this variable shall be FALSE.

14.8.61 pdelayRespSendDisabled

The value is equal to the value of the per-PTP Port variable pdelayRespSendDisabled (see 11.2.20.2.6). If its value is TRUE, Pdelay_Resp and Pdelay_Resp_Follow_Up messages are not transmitted by the PTP Port in response to Pdelay_Req message. The default value for this variable shall be FALSE.

17. YANG Data Model

17.2 IEEE 802.1AS YANG models

Replace Figure 17-3 with the following:

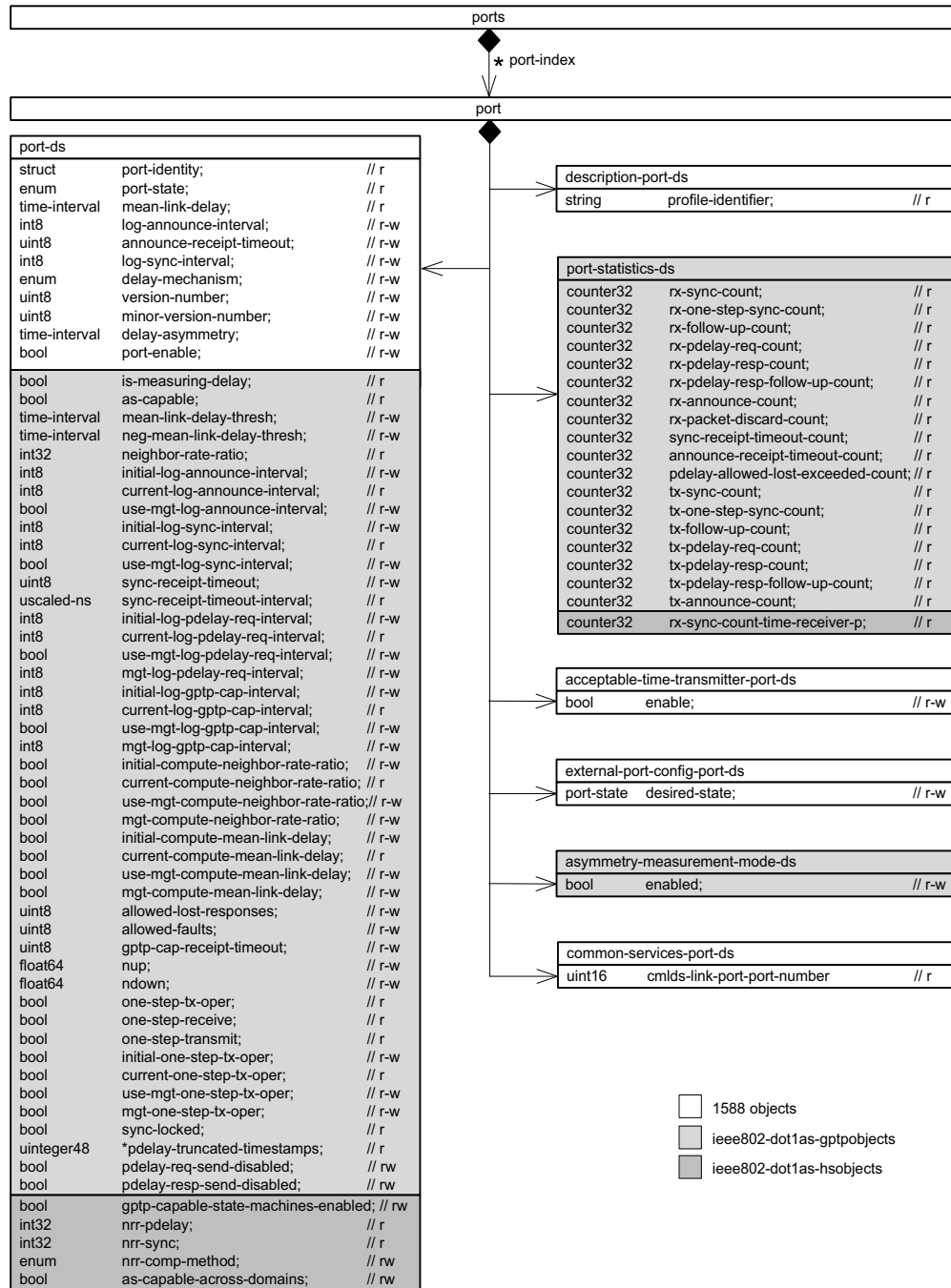


Figure 17-3—PTP Port detail

1 17.5 YANG schema tree definitions

2 17.5.1 ITree diagram for ieee802-dot1as-gptp.yang

3 *Replace the tree diagram for ieee802-dot1as-gptp.yang as follows:*

```
4 module: ieee802-dot1as-gptp
5
6   augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:default-ds:
7     +--ro gm-capable?                boolean
8     +--ro current-utc-offset?        int16
9     +--ro current-utc-offset-valid?  boolean
10    +--ro leap59?                    boolean
11    +--ro leap61?                    boolean
12    +--ro time-traceable?             boolean
13    +--ro frequency-traceable?        boolean
14    +--ro ptp-timescale?              boolean
15    +--ro time-source?                identityref
16  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:current-ds:
17    +--ro last-gm-phase-change?       scaled-ns
18    +--ro last-gm-freq-change?        float64
19    +--ro gm-timebase-indicator?      uint16
20    +--ro gm-change-count?            yang:counter32
21    +--ro time-of-last-gm-change?     yang:timestamp
22    +--ro time-of-last-phase-change?  yang:timestamp
23    +--ro time-of-last-freq-change?   yang:timestamp
24  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:parent-ds:
25    +--ro cumulative-rate-ratio?      int32
26  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port/
27  ptp-tt:port-ds:
28    +--ro is-measuring-delay?          boolean
29    +--ro as-capable?                  boolean
30    +--rw mean-link-delay-thresh?      ptp-tt:time-interval
31    +--rw neg-mean-link-delay-thresh?  ptp-tt:time-interval
32    +--ro neighbor-rate-ratio?         int32
33    +--rw initial-log-announce-interval? int8
34    +--ro current-log-announce-interval? int8
35    +--rw use-mgt-log-announce-interval? boolean
36    +--rw initial-log-sync-interval?   int8
37    +--ro current-log-sync-interval?   int8
38    +--rw use-mgt-log-sync-interval?   boolean
39    +--rw sync-receipt-timeout?        uint8
40    +--ro sync-receipt-timeout-interval? uscaled-ns
41    +--rw initial-log-pdelay-req-interval? int8
42    +--ro current-log-pdelay-req-interval? int8
43    +--rw use-mgt-log-pdelay-req-interval? boolean
44    +--rw mgt-log-pdelay-req-interval?  int8
45    +--rw initial-log-gptp-cap-interval? int8
46    +--ro current-log-gptp-cap-interval? int8
47    +--rw use-mgt-log-gptp-cap-interval? boolean
48    +--rw mgt-log-gptp-cap-interval?    int8
49    +--rw initial-compute-neighbor-rate-ratio? boolean
50    +--ro current-compute-neighbor-rate-ratio? boolean
51    +--rw use-mgt-compute-neighbor-rate-ratio? boolean
52    +--rw mgt-compute-neighbor-rate-ratio? boolean
53    +--rw initial-compute-mean-link-delay? boolean
54    +--ro current-compute-mean-link-delay? boolean
55    +--rw use-mgt-compute-mean-link-delay? boolean
```

```

1      1      +--rw mgt-compute-mean-link-delay?          boolean
2      2      +--rw allowed-lost-responses?              uint8
3      3      +--rw allowed-faults?                      uint8
4      4      +--rw gptp-cap-receipt-timeout?            uint8
5      5      +--rw nup?                                  float64
6      6      +--rw ndown?                                float64
7      7      +--ro one-step-tx-oper?                    boolean
8      8      +--ro one-step-receive?                    boolean
9      9      +--ro one-step-transmit?                   boolean
10     10     +--rw initial-one-step-tx-oper?             boolean
11     11     +--ro current-one-step-tx-oper?             boolean
12     12     +--rw use-mgt-one-step-tx-oper?             boolean
13     13     +--rw mgt-one-step-tx-oper?                 boolean
14     14     +--ro sync-locked?                          boolean
15     15     +--ro pdelay-truncated-timestamps*          uinteger48
16     16     +--rw pdelay-req-send-disabled?            boolean
17     17     +--rw pdelay-resp-send-disabled?            boolean
18     18     augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
19     19         +--rw port-statistics-ds
20     20             +--ro rx-sync-count?                 yang:counter32
21     21             +--ro rx-one-step-sync-count?        yang:counter32
22     22             +--ro rx-follow-up-count?            yang:counter32
23     23             +--ro rx-pdelay-req-count?           yang:counter32
24     24             +--ro rx-pdelay-resp-count?          yang:counter32
25     25             +--ro rx-pdelay-resp-follow-up-count? yang:counter32
26     26             +--ro rx-announce-count?             yang:counter32
27     27             +--ro rx-packet-discard-count?       yang:counter32
28     28             +--ro sync-receipt-timeout-count?    yang:counter32
29     29             +--ro announce-receipt-timeout-count? yang:counter32
30     30             +--ro pdelay-allowed-lost-exceeded-count? yang:counter32
31     31             +--ro tx-sync-count?                 yang:counter32
32     32             +--ro tx-one-step-sync-count?        yang:counter32
33     33             +--ro tx-follow-up-count?            yang:counter32
34     34             +--ro tx-pdelay-req-count?           yang:counter32
35     35             +--ro tx-pdelay-resp-count?          yang:counter32
36     36             +--ro tx-pdelay-resp-follow-up-count? yang:counter32
37     37             +--ro tx-announce-count?             yang:counter32
38     38     augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
39     39         +--rw asymmetry-measurement-mode-ds
40     40         +--rw enabled? boolean
41     41         augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmls/ptp-tt:ports/ptp-
42     42 tt:port/ptp-tt:link-port-ds:
43     43             +--ro cmls-link-port-enabled?         boolean
44     44             +--ro is-measuring-delay?              boolean
45     45             +--ro as-capable-across-domains?      boolean
46     46             +--rw mean-link-delay-thresh?         ptp-tt:time-interval
47     47             +--rw neg-mean-link-delay-thresh?     ptp-tt:time-interval
48     48             +--rw initial-log-pdelay-req-interval? int8
49     49             +--ro current-log-pdelay-req-interval? int8
50     50             +--rw use-mgt-log-pdelay-req-interval? boolean
51     51             +--rw mgt-log-pdelay-req-interval?     int8
52     52             +--rw initial-compute-neighbor-rate-ratio? boolean
53     53             +--ro current-compute-neighbor-rate-ratio? boolean
54     54             +--rw use-mgt-compute-neighbor-rate-ratio? boolean
55     55             +--rw mgt-compute-neighbor-rate-ratio? boolean
56     56             +--rw initial-compute-mean-link-delay? boolean
57     57             +--ro current-compute-mean-link-delay? boolean
58     58             +--rw use-mgt-compute-mean-link-delay? boolean
59     59             +--rw mgt-compute-mean-link-delay?    boolean

```

```

1   +--rw allowed-lost-responses?                uint8
2   +--rw allowed-faults?                        uint8
3   +--ro pdelay-truncated-timestamps*          uinteger48
4   augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmds/ptp-tt:ports/ptp-
5 tt:port:
6   +--rw port-statistics-ds
7       +--ro rx-pdelay-req-count?               yang:counter32
8       +--ro rx-pdelay-resp-count?              yang:counter32
9       +--ro rx-pdelay-resp-follow-up-count?    yang:counter32
10      +--ro rx-packet-discard-count?            yang:counter32
11      +--ro pdelay-allowed-lost-exceeded-count? yang:counter32
12      +--ro tx-pdelay-req-count?                yang:counter32
13      +--ro tx-pdelay-resp-count?              yang:counter32
14      +--ro tx-pdelay-resp-follow-up-count?    yang:counter32
15      augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmds/ptp-tt:ports/ptp-
16 tt:port:
17      +--rw asymmetry-measurement-mode-ds
18      +--rw enabled?    boolean
19

```

17.6 YANG modules^{1 2}

17.6.1 Module `ieee802-dot1as-gtp.yang`

Replace the YANG module for `ieee802-dot1as-gtp.yang` as follows:

```

23 module ieee802-dot1as-gtp {
24   yang-version 1.1;
25   namespace "urn:ieee:std:802.1AS:yang:ieee802-dot1as-gtp";
26   prefix dot1as-gtp;
27
28   import ietf-yang-types {
29     prefix yang;
30   }
31   import ieee1588-ptp-tt {
32     prefix ptp-tt;
33   }
34
35   organization
36     "IEEE 802.1 Working Group";
37   contact
38     "WG-URL: http://ieee802.org/1/
39     WG-EMail: stds-802-1-l@ieee.org
40
41     Contact: IEEE 802.1 Working Group Chair
42             Postal: C/O IEEE 802.1 Working Group
43             IEEE Standards Association
44             445 Hoes Lane
45             Piscataway, NJ 08854
46             USA
47
48     E-mail: stds-802-1-chairs@ieee.org";
49   description

```

¹Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

²An ASCII version of the YANG modules are attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.

```

1      "Management objects that control timing and synchronization for time
2      sensitive applications, as specified in clause 14 of IEEE Std
3      802.1AS-2020-REV.
4
5      Copyright (C) IEEE (2025). This version of this YANG module is part
6      of IEEE Std 802.1AS-2020-REV; see the standard itself for full legal
7      notices.";
8
9      revision 2025-03-06 {
10         description
11             "Published as part of IEEE Std 802.1AS-2020-REV.";
12         reference
13             "IEEE Std 802.1AS - Timing and Synchronization for Time-Sensitive
14             Applications: IEEE Std 802.1AS-2020-REV.
15             IEEE Std 1588 - IEEE Standard for a Precision Clock Synchronization
16             Protocol for Networked Measurement and Control Systems: IEEE Std
17             1588-2019, IEEE Std 1588g-2022, IEEE Std 1588e-2024.";
18     }
19
20     typedef scaled-ns {
21         type string {
22             pattern '[0-9A-F]{2}(-[0-9A-F]{2}){11}';
23         }
24         description
25             "The IEEE Std 802.1AS ScaledNs type represents signed values of
26             time and time interval in units of 2^16 ns, as a signed 96-bit
27             integer. Each of the 12 octets is represented as a pair of
28             hexadecimal characters, using uppercase for a letter. Octets are
29             separated by a dash character. The most significant octet is first.";
30         reference
31             "6.4.3.1 of IEEE Std 802.1AS";
32     }
33
34     typedef unscaled-ns {
35         type string {
36             pattern '[0-9A-F]{2}(-[0-9A-F]{2}){11}';
37         }
38         description
39             "The IEEE Std 802.1AS UScaledNs type represents unsigned values of
40             time and time interval in units of 2^16 ns, as an unsigned 96-bit
41             integer. Each of the 12 octets is represented as a pair of
42             hexadecimal characters, using uppercase for a letter. Octets are
43             separated by a dash character. The most significant octet is first.";
44         reference
45             "6.4.3.2 of IEEE Std 802.1AS";
46     }
47
48     typedef float64 {
49         type string {
50             pattern '[0-9A-F]{2}(-[0-9A-F]{2}){7}';
51         }
52         description
53             "The IEEE Std 802.1AS Float64 type represents IEEE Std 754
54             binary64. Each of the 8 octets is represented as a pair of
55             hexadecimal characters, using uppercase for a letter. Octets are
56             separated by a dash character. The most significant octet is first.";
57         reference
58             "6.4.2 of IEEE Std 802.1AS";
59     }

```

```

1
2 typedef uinteger48 {
3     type uint64 {
4         range "0..281474976710655";
5     }
6     description
7         "48-bit unsigned integer data type.";
8     reference
9         "6.4.2 of IEEE Std 802.1AS";
10 }
11
12 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
13     + "/ptp-tt:default-ds" {
14     description
15         "Augment IEEE Std 1588 defaultDS.";
16     leaf gm-capable {
17         type boolean;
18         config false;
19         description
20             "The value is true if the time-aware system is capable of being a
21             Grandmaster, and false if the time-aware system is not capable of
22             being a Grandmaster.";
23         reference
24             "14.2.7 of IEEE Std 802.1AS";
25     }
26     leaf current-utc-offset {
27         when "../current-utc-offset-valid='true'";
28         type int16;
29         config false;
30         description
31             "Offset from UTC (TAI - UTC). The offset is in units of seconds.
32             This leaf applies to the ClockTimeTransmitter entity (i.e., local
33             only, unrelated to a remote GM).";
34         reference
35             "14.2.8 of IEEE Std 802.1AS";
36     }
37     leaf current-utc-offset-valid {
38         type boolean;
39         config false;
40         description
41             "The value of current-utc-offset-valid shall be true if the value
42             of current-utc-offset is known to be correct, otherwise it shall
43             be false. This leaf applies to the ClockTimeTransmitter entity
44             (i.e., local only, unrelated to a remote GM).";
45         reference
46             "14.2.9 of IEEE Std 802.1AS";
47     }
48     leaf leap59 {
49         type boolean;
50         config false;
51         description
52             "If the timescale is PTP, a true value for leap59 shall indicate
53             that the last minute of the current UTC day contains 59 seconds.
54             If the timescale is not PTP, the value shall be false. This leaf
55             applies to the ClockTimeTransmitter entity (i.e., local only,
56             unrelated to a remote GM).";
57         reference
58             "14.2.10 of IEEE Std 802.1AS";
59     }

```

```

1      1      leaf leap61 {
2      2          type boolean;
3      3          config false;
4      4          description
5      5              "If the timescale is PTP, a true value for leap61 shall indicate
6      6              that the last minute of the current UTC day contains 61 seconds.
7      7              If the timescale is not PTP, the value shall be false. This leaf
8      8              applies to the ClockTimeTransmitter entity (i.e., local only,
9      9              unrelated to a remote GM).";
10     10         reference
11     11             "14.2.11 of IEEE Std 802.1AS";
12     12     }
13     13     leaf time-traceable {
14     14         type boolean;
15     15         config false;
16     16         description
17     17             "The value of time-traceable shall be true if the timescale is
18     18             traceable to a primary reference; otherwise, the value shall be
19     19             false. This leaf applies to the ClockTimeTransmitter entity
20     20             (i.e., local only, unrelated to a remote GM).";
21     21         reference
22     22             "14.2.12 of IEEE Std 802.1AS";
23     23     }
24     24     leaf frequency-traceable {
25     25         type boolean;
26     26         config false;
27     27         description
28     28             "The value of frequency-traceable shall be true if the frequency
29     29             determining the timescale is traceable to a primary reference;
30     30             otherwise, the value shall be false. This leaf applies to the
31     31             ClockTimeTransmitter entity (i.e., local only, unrelated to a
32     32             remote GM).";
33     33         reference
34     34             "14.2.13 of IEEE Std 802.1AS";
35     35     }
36     36     leaf ptp-timescale {
37     37         type boolean;
38     38         config false;
39     39         description
40     40             "If ptp-timescale is true, the timescale of the
41     41             ClockTimeTransmitter entity is PTP, which is the elapsed time
42     42             since the PTP epoch measured using the second defined by
43     43             International Atomic Time (TAI). If ptp-timescale is false, the
44     44             timescale of the ClockTimeTransmitter entity is ARB, which is the
45     45             elapsed time since an arbitrary epoch. This leaf applies to the
46     46             ClockTimeTransmitter entity (i.e., local only, unrelated to a
47     47             remote GM).";
48     48         reference
49     49             "14.2.14 of IEEE Std 802.1AS";
50     50     }
51     51     leaf time-source {
52     52         type identityref {
53     53             base ptp-tt:time-source;
54     54         }
55     55         config false;
56     56         description
57     57             "The source of time used by the Grandmaster Clock. This leaf
58     58             applies to the ClockTimeTransmitter entity (i.e., local only,
59     59             unrelated to a remote GM).";

```

```

1      reference
2      "14.2.15 of IEEE Std 802.1AS";
3  }
4  }
5
6  augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
7      + "/ptp-tt:current-ds" {
8      description
9      "Augment IEEE Std 1588 currentDS.";
10     leaf last-gm-phase-change {
11         type scaled-ns;
12         config false;
13         description
14         "Phase change that occurred on the most recent change in either
15         the Grandmaster PTP Instance or gm-timebase-indicator leaf.";
16         reference
17         "14.3.4 of IEEE Std 802.1AS";
18     }
19     leaf last-gm-freq-change {
20         type float64;
21         config false;
22         description
23         "Frequency change that occurred on the most recent change in
24         either the Grandmaster PTP Instance or gm-timebase-indicator
25         leaf.";
26         reference
27         "14.3.5 of IEEE Std 802.1AS";
28     }
29     leaf gm-timebase-indicator {
30         type uint16;
31         config false;
32         description
33         "The timeBaseIndicator of the current Grandmaster PTP Instance.";
34         reference
35         "14.3.6 of IEEE Std 802.1AS";
36     }
37     leaf gm-change-count {
38         type yang:counter32;
39         config false;
40         description
41         "This statistics counter tracks the number of times the
42         Grandmaster PTP Instance has changed in a gPTP domain.";
43         reference
44         "14.3.7 of IEEE Std 802.1AS";
45     }
46     leaf time-of-last-gm-change {
47         type yang:timestamp;
48         config false;
49         description
50         "System time when the most recent Grandmaster Clock change
51         occurred in a gPTP domain. This leaf's type is YANG timestamp,
52         which is based on system time. System time is an unsigned integer
53         in units of 10 milliseconds, using an epoch defined by the
54         implementation (typically time of boot-up).";
55         reference
56         "14.3.8 of IEEE Std 802.1AS";
57     }
58     leaf time-of-last-phase-change {
59         type yang:timestamp;

```

```

1      1      config false;
2      2      description
3      3          "System time when the most recent change in Grandmaster Clock
4      4          phase occurred. This leaf's type is YANG timestamp, which is
5      5          based on system time. System time is an unsigned integer in units
6      6          of 10 milliseconds, using an epoch defined by the implementation
7      7          (typically time of boot-up).";
8      8      reference
9      9          "14.3.9 of IEEE Std 802.1AS";
10     10     }
11     11     leaf time-of-last-freq-change {
12     12         type yang:timestamp;
13     13         config false;
14     14         description
15     15             "System time when the most recent change in Grandmaster Clock
16     16             frequency occurred. This leaf's type is YANG timestamp, which is
17     17             based on system time. System time is an unsigned integer in units
18     18             of 10 milliseconds, using an epoch defined by the implementation
19     19             (typically time of boot-up).";
20     20         reference
21     21             "14.3.10 of IEEE Std 802.1AS";
22     22     }
23     23 }
24     24
25     25 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
26     26     + "/ptp-tt:parent-ds" {
27     27         description
28     28             "Augment IEEE Std 1588 parentDS.";
29     29         leaf cumulative-rate-ratio {
30     30             type int32;
31     31             config false;
32     32             description
33     33                 "Estimate of the ratio of the frequency of the Grandmaster Clock
34     34                 to the frequency of the LocalClock entity of this PTP Instance.
35     35                 cumulative-rate-ratio is expressed as the fractional frequency
36     36                 offset multiplied by 2^41, i.e., the quantity (rateRatio -
37     37                 1.0)(2^41).";
38     38             reference
39     39                 "14.4.3 of IEEE Std 802.1AS";
40     40         }
41     41     }
42     42
43     43 augment "/ptp-tt:ptp"
44     44     + "/ptp-tt:instances"
45     45     + "/ptp-tt:instance"
46     46     + "/ptp-tt:ports"
47     47     + "/ptp-tt:port"
48     48     + "/ptp-tt:port-ds" {
49     49         description
50     50             "Augment IEEE Std 1588 portDS.
51     51
52     52             14.10.4 of IEEE Std 802.1AS specifies ptpPortEnabled
53     53             (ptp-port-enabled), which is provided in YANG as the semantically
54     54             equivalent node in ieee1588-ptp-tt named port-enable (in port-ds).
55     55
56     56             14.10.15 of IEEE Std 802.1AS specifies
57     57             mgtSettableLogAnnounceInterval (mgt-log-announce-interval), which
58     58             is provided in YANG as the semantically equivalent node in
59     59             ieee1588-ptp-tt named log-announce-interval (in port-ds). In the

```

```

1      context of IEEE Std 802.1AS, log-announce-interval cannot be used
2      unless use-mgt-log-announce-interval is true.
3
4      14.10.20 of IEEE Std 802.1AS specifies mgtSettableLogSyncInterval
5      (mgt-log-sync-interval), which is provided in YANG as the
6      semantically equivalent node in ieee1588-ptp-tt named
7      log-sync-interval (in port-ds). In the context of IEEE Std 802.1AS,
8      log-sync-interval cannot be used unless use-mgt-log-sync-interval
9      is true.";
10     leaf is-measuring-delay {
11         type boolean;
12         config false;
13         description
14             "Boolean that is true if the port is measuring PTP Link
15             propagation delay.";
16         reference
17             "14.10.6 of IEEE Std 802.1AS";
18     }
19     leaf as-capable {
20         type boolean;
21         config false;
22         description
23             "Boolean that is true if and only if it is determined that this
24             PTP Instance and the PTP Instance at the other end of the link
25             attached to this port can interoperate with each other via the
26             IEEE Std 802.1AS protocol.";
27         reference
28             "10.2.5.1 of IEEE Std 802.1AS
29             14.10.7 of IEEE Std 802.1AS";
30     }
31     leaf mean-link-delay-thresh {
32         type ptp-tt:time-interval;
33         description
34             "Propagation time threshold for mean-link-delay, above which a
35             port is not considered capable of participating in the IEEE Std
36             802.1AS protocol.";
37         reference
38             "14.10.9 of IEEE Std 802.1AS";
39     }
40     leaf neg-mean-link-delay-thresh {
41         type ptp-tt:time-interval;
42         description
43             "The negative propagation time threshold for mean-link-delay,
44             below which a port is not considered capable of participating
45             in the IEEE Std 802.1AS protocol.";
46         reference
47             "14.10.10 of IEEE Std 802.1AS";
48     }
49     leaf neighbor-rate-ratio {
50         type int32;
51         config false;
52         description
53             "Estimate of the ratio of the frequency of the LocalClock entity
54             of the PTP Instance at the other end of the link attached to this
55             PTP Port, to the frequency of the LocalClock entity of this PTP
56             Instance. neighbor-rate-ratio is expressed as the fractional
57             frequency offset multiplied by 2^41, i.e., the quantity
58             (rateRatio - 1.0)(2^41).";
59         reference

```

```

1      1      "14.10.12 of IEEE Std 802.1AS";
2      2      }
3      3      leaf initial-log-announce-interval {
4      4          type int8;
5      5          description
6      6              "When use-mgt-log-announce-interval is false (i.e., change with
7      7                  Signaling message), this is the the logarithm to base 2 of the
8      8                      announce interval used when the port is initialized.";
9      9          reference
10     10              "14.10.13 of IEEE Std 802.1AS";
11     11      }
12     12      leaf current-log-announce-interval {
13     13          type int8;
14     14          config false;
15     15          description
16     16              "Logarithm to base 2 of the current announce interval.";
17     17          reference
18     18              "14.10.14 of IEEE Std 802.1AS";
19     19      }
20     20      leaf use-mgt-log-announce-interval {
21     21          type boolean;
22     22          description
23     23              "Boolean that determines the source of the announce interval. If
24     24                  the value is true, the announce interval
25     25                      (current-log-announce-interval) is set equal to the value of
26     26                      mgt-log-announce-interval. If the value is false, the announce
27     27                      interval is determined by the AnnounceIntervalSetting state
28     28                      machine (i.e., changed with Signaling message).";
29     29          reference
30     30              "14.10.15 of IEEE Std 802.1AS";
31     31      }
32     32      leaf initial-log-sync-interval {
33     33          type int8;
34     34          description
35     35              "When use-mgt-log-sync-interval is false (i.e., change with
36     36                  Signaling message), this is the the logarithm to base 2 of the
37     37                      sync interval used when the port is initialized.";
38     38          reference
39     39              "14.10.17 of IEEE Std 802.1AS";
40     40      }
41     41      leaf current-log-sync-interval {
42     42          type int8;
43     43          config false;
44     44          description
45     45              "Logarithm to base 2 of the current sync interval.";
46     46          reference
47     47              "14.10.18 of IEEE Std 802.1AS";
48     48      }
49     49      leaf use-mgt-log-sync-interval {
50     50          type boolean;
51     51          description
52     52              "Boolean that determines the source of the sync interval. If the
53     53                  value is true, the sync interval (current-log-sync-interval) is
54     54                      set equal to the value of mgt-log-sync-interval. If the value is
55     55                      false, the sync interval is determined by the SyncIntervalSetting
56     56                      state machine (i.e., changed with Signaling message).";
57     57          reference
58     58              "14.10.19 of IEEE Std 802.1AS";
59     59      }

```

```

1  leaf sync-receipt-timeout {
2      type uint8;
3      description
4          "Number of sync intervals that a timeReceiver port waits without
5          receiving synchronization information, before assuming that the
6          timeTransmitter is no longer transmitting synchronization
7          information and that the BTCA needs to be run, if appropriate.";
8      reference
9          "14.10.21 of IEEE Std 802.1AS";
10 }
11 leaf sync-receipt-timeout-interval {
12     type uscaled-ns;
13     config false;
14     description
15         "Time interval after which sync receipt timeout occurs if
16         time-synchronization information has not been received during the
17         interval.";
18     reference
19         "14.10.22 of IEEE Std 802.1AS";
20 }
21 leaf initial-log-pdelay-req-interval {
22     type int8;
23     description
24         "When use-mgt-log-pdelay-req-interval is false (i.e., change with
25         Signaling message), this is the the logarithm to base 2 of the
26         Pdelay_Req transmit interval used when the port is initialized.";
27     reference
28         "14.10.23 of IEEE Std 802.1AS";
29 }
30 leaf current-log-pdelay-req-interval {
31     type int8;
32     config false;
33     description
34         "Logarithm to base 2 of the current Pdelay_Req transmit interval.";
35     reference
36         "14.10.24 of IEEE Std 802.1AS";
37 }
38 leaf use-mgt-log-pdelay-req-interval {
39     type boolean;
40     description
41         "Boolean that determines the source of the Pdelay_Req transmit
42         interval. If the value is true, the Pdelay_Req transmit interval
43         (current-log-pdelay-req-interval) is set equal to the value of
44         mgt-log-pdelay-req-interval. If the value is false, the
45         Pdelay_Req transmit interval is determined by the
46         LinkDelayIntervalSetting state machine (i.e., changed with
47         Signaling message).";
48     reference
49         "14.10.25 of IEEE Std 802.1AS";
50 }
51 leaf mgt-log-pdelay-req-interval {
52     type int8;
53     description
54         "Logarithm to base 2 of the Pdelay_Req transmit interval, used if
55         use-mgt-log-pdelay-req-interval is true. This value is not used
56         if use-mgt-log-pdelay-req-interval is false.";
57     reference
58         "14.10.26 of IEEE Std 802.1AS";
59 }

```

```

1      1      leaf initial-log-gtp-cap-interval {
2      2          type int8;
3      3          description
4      4              "When use-mgt-log-gtp-cap-interval is false (i.e., change with
5      5                  Signaling message), this is the the logarithm to base 2 of the
6      6                  gTP capable message interval used when the port is initialized.";
7      7          reference
8      8              "14.10.27 of IEEE Std 802.1AS";
9      9      }
10     10     leaf current-log-gtp-cap-interval {
11     11         type int8;
12     12         config false;
13     13         description
14     14             "Logarithm to base 2 of the current gTP capable message
15     15                 interval.";
16     16         reference
17     17             "14.10.28 of IEEE Std 802.1AS";
18     18     }
19     19     leaf use-mgt-log-gtp-cap-interval {
20     20         type boolean;
21     21         description
22     22             "Boolean that determines the source of the gTP capable message
23     23                 interval. If the value is true, the gTP capable message interval
24     24                 (current-log-gtp-cap-interval) is set equal to the value of
25     25                 mgt-gtp-cap-req-interval. If the value is false, the gTP
26     26                 capable message interval is determined by the
27     27                 GtpCapableMessageIntervalSetting state machine (i.e., changed
28     28                 with Signaling message).";
29     29         reference
30     30             "14.10.29 of IEEE Std 802.1AS";
31     31     }
32     32     leaf mgt-log-gtp-cap-interval {
33     33         type int8;
34     34         description
35     35             "Logarithm to base 2 of the gTP capable message interval, used
36     36                 if use-mgt-log-gtp-cap-interval is true. This value is not used
37     37                 if use-mgt-log-pdelay-req-interval is false.";
38     38         reference
39     39             "14.10.30 of IEEE Std 802.1AS";
40     40     }
41     41     leaf initial-compute-neighbor-rate-ratio {
42     42         type boolean;
43     43         description
44     44             "When use-mgt-compute-neighbor-rate-ratio is false (i.e., change
45     45                 with Signaling message), this is the initial value of
46     46                 computeNeighborRateRatio.";
47     47         reference
48     48             "14.10.31 of IEEE Std 802.1AS";
49     49     }
50     50     leaf current-compute-neighbor-rate-ratio {
51     51         type boolean;
52     52         config false;
53     53         description
54     54             "Current value of computeNeighborRateRatio.";
55     55         reference
56     56             "14.10.32 of IEEE Std 802.1AS";
57     57     }
58     58     leaf use-mgt-compute-neighbor-rate-ratio {
59     59         type boolean;

```

```

1      description
2      "Boolean that determines the source of computeNeighborRateRatio..
3      If the value is true, computeNeighborRateRatio is set equal to
4      the value of mgt-compute-neighbor-rate-ratio. If the value is
5      false, computeNeighborRateRatio is determined by the
6      LinkDelayIntervalSetting state machine (i.e., changed with
7      Signaling message).";
8      reference
9      "14.10.33 of IEEE Std 802.1AS";
10     }
11     leaf mgt-compute-neighbor-rate-ratio {
12         type boolean;
13         description
14         "Value of computeNeighborRateRatio, used if
15         use-mgt-compute-neighbor-rate-ratio is true. This value is not
16         used if use-mgt-compute-neighbor-rate-ratio is false.";
17         reference
18         "14.10.34 of IEEE Std 802.1AS";
19     }
20     leaf initial-compute-mean-link-delay {
21         type boolean;
22         description
23         "When use-mgt-compute-mean-link-delay is false (i.e., change with
24         Signaling message), this is the initial value of
25         computeMeanLinkDelay.";
26         reference
27         "14.10.35 of IEEE Std 802.1AS";
28     }
29     leaf current-compute-mean-link-delay {
30         type boolean;
31         config false;
32         description
33         "Current value of computeMeanLinkDelay.";
34         reference
35         "14.10.36 of IEEE Std 802.1AS";
36     }
37     leaf use-mgt-compute-mean-link-delay {
38         type boolean;
39         description
40         "Boolean that determines the source of computeMeanLinkDelay. If
41         the value is true, computeMeanLinkDelay is set equal to the value
42         of mgt-compute-mean-link-delay. If the value is false,
43         computeMeanLinkDelay is determined by the
44         LinkDelayIntervalSetting state machine (i.e., changed with
45         Signaling message).";
46         reference
47         "14.10.37 of IEEE Std 802.1AS";
48     }
49     leaf mgt-compute-mean-link-delay {
50         type boolean;
51         description
52         "Value of computeMeanLinkDelay, used if
53         use-mgt-compute-mean-link-delay is true. This value is not used
54         if use-mgt-compute-mean-link-delay is false.";
55         reference
56         "14.10.38 of IEEE Std 802.1AS";
57     }
58     leaf allowed-lost-responses {
59         type uint8;

```

```

1      1      description
2      2      "Number of Pdelay_Req messages for which a valid response is not
3      3      received, above which a port is considered to not be exchanging
4      4      peer delay messages with its neighbor.";
5      5      reference
6      6      "14.10.39 of IEEE Std 802.1AS";
7      7      }
8      8      leaf allowed-faults {
9      9      type uint8;
10     10     description
11     11     "Number of faults above which asCapable is set to false.";
12     12     reference
13     13     "14.10.40 of IEEE Std 802.1AS";
14     14     }
15     15     leaf gptp-cap-receipt-timeout {
16     16     type uint8;
17     17     description
18     18     "Number of transmission intervals that a port waits without
19     19     receiving the gPTP capable TLV, before assuming that the neighbor
20     20     port is no longer invoking the gPTP protocol.";
21     21     reference
22     22     "14.10.41 of IEEE Std 802.1AS";
23     23     }
24     24     leaf nup {
25     25     type float64;
26     26     description
27     27     "For an OLT port of an IEEE Std 802.3 EPON link, this value is
28     28     the effective index of refraction for the EPON upstream
29     29     wavelength light of the optical path.";
30     30     reference
31     31     "14.10.43 of IEEE Std 802.1AS";
32     32     }
33     33     leaf ndown {
34     34     type float64;
35     35     description
36     36     "For an OLT port of an IEEE 802.3 EPON link, this value is the
37     37     effective index of refraction for the EPON downstream wavelength
38     38     light of the optical path.";
39     39     reference
40     40     "14.10.44 of IEEE Std 802.1AS";
41     41     }
42     42     leaf one-step-tx-oper {
43     43     type boolean;
44     44     config false;
45     45     description
46     46     "This value is true if the port is sending one-step Sync
47     47     messages, and false if the port is sending two-step Sync and
48     48     Follow-Up messages.";
49     49     reference
50     50     "14.10.45 of IEEE Std 802.1AS";
51     51     }
52     52     leaf one-step-receive {
53     53     type boolean;
54     54     config false;
55     55     description
56     56     "This value is true if the port is capable of receiving and
57     57     processing one-step Sync messages.";
58     58     reference
59     59     "14.10.46 of IEEE Std 802.1AS";

```

```

1   }
2   leaf one-step-transmit {
3       type boolean;
4       config false;
5       description
6           "This value is true if the port is capable of transmitting
7           one-step Sync messages.";
8       reference
9           "14.10.47 of IEEE Std 802.1AS";
10  }
11  leaf initial-one-step-tx-oper {
12      type boolean;
13      description
14          "When use-mgt-one-step-tx-oper is false (i.e., change with
15          Signaling message), this is the initial value of
16          current-one-step-tx-oper.";
17      reference
18          "14.10.48 of IEEE Std 802.1AS";
19  }
20  leaf current-one-step-tx-oper {
21      type boolean;
22      config false;
23      description
24          "This value is true if the port is configured to transmit
25          one-step Sync messages, either via management
26          (mgt-one-step-tx-oper) or Signaling. If both
27          current-one-step-tx-oper and one-step-transmit are true, the port
28          transmits one-step Sync messages (i.e., one-step-tx-oper true).";
29      reference
30          "14.10.49 of IEEE Std 802.1AS";
31  }
32  leaf use-mgt-one-step-tx-oper {
33      type boolean;
34      description
35          "Boolean that determines the source of current-one-step-tx-oper.
36          If the value is true, current-one-step-tx-oper is set equal to
37          the value of mgt-one-step-tx-oper. If the value is false,
38          current-one-step-tx-oper is determined by the
39          OneStepTxOperSetting state machine (i.e., changed with Signaling
40          message).";
41      reference
42          "14.10.50 of IEEE Std 802.1AS";
43  }
44  leaf mgt-one-step-tx-oper {
45      type boolean;
46      description
47          "If use-mgt-one-step-tx-oper is true, current-one-step-tx-oper is
48          set equal to this value. This value is not used if
49          use-mgt-one-step-tx-oper is false.";
50      reference
51          "14.10.51 of IEEE Std 802.1AS";
52  }
53  leaf sync-locked {
54      type boolean;
55      config false;
56      description
57          "This value is true if the port will transmit a Sync as soon as
58          possible after the timeReceiver port receives a Sync message.";
59      reference

```

```

1      "14.10.52 of IEEE Std 802.1AS";
2    }
3    leaf-list pdelay-truncated-timestamps {
4      type uinteger48;
5      config false;
6      description
7        "For full-duplex IEEE Std 802.3 media, and CSN media that use the
8        peer-to-peer delay mechanism to measure path delay, the values of
9        the four elements of this leaf-list correspond to the timestamps
10       t1, t2, t3, and t4, listed in that order. Each timestamp is
11       expressed in units of 2-16 ns (i.e., the value of each array
12       element is equal to the remainder obtained upon dividing the
13       respective timestamp, expressed in units of 2-16 ns, by 248).
14       At any given time, the timestamp values stored in the array are
15       for the same, and most recently completed, peer delay message
16       exchange. For each timestamp, only 48-bits are valid (the upper
17       16-bits are always zero).";
18     reference
19       "14.10.53 of IEEE Std 802.1AS";
20   }
21   leaf pdelay-req-send-disabled {
22     type boolean;
23     description
24       "A boolean that is administratively set to TRUE
25       if Pdelay_Req messages are not transmitted by this port.
26       The default value for this variable shall be FALSE.";
27     reference
28       "11.2.19.2.14 of IEEE Std 802.1ASds";
29   }
30   leaf pdelay-resp-send-disabled {
31     type boolean;
32     description
33       "A boolean that is administratively set to TRUE
34       if Pdelay_Resp messages are not transmitted by this port.
35       The default value for this variable shall be FALSE.";
36     reference
37       "11.2.20.2.6 of IEEE Std 802.1ASds";
38   }
39 }
40
41 augment "/ptp-tt:ptp"
42   + "/ptp-tt:instances"
43   + "/ptp-tt:instance"
44   + "/ptp-tt:ports"
45   + "/ptp-tt:port" {
46   description
47     "Augment to add port-statistics-ds to IEEE Std 1588 PTP Port.";
48   container port-statistics-ds {
49     description
50       "Provides counters associated with the port of the PTP Instance.";
51     reference
52       "14.12 of IEEE Std 802.1AS";
53     leaf rx-sync-count {
54       type yang:counter32;
55       config false;
56       description
57         "Counter that increments every time synchronization information
58         is received.";
59       reference

```

```

1      "14.12.2 of IEEE Std 802.1AS";
2  }
3  leaf rx-one-step-sync-count {
4      type yang:counter32;
5      config false;
6      description
7          "Counter that increments every time a one-step Sync message is
8          received.";
9      reference
10         "14.12.3 of IEEE Std 802.1AS";
11 }
12 leaf rx-follow-up-count {
13     type yang:counter32;
14     config false;
15     description
16         "Counter that increments every time a Follow_Up message is
17         received.";
18     reference
19         "14.12.4 of IEEE Std 802.1AS";
20 }
21 leaf rx-pdelay-req-count {
22     type yang:counter32;
23     config false;
24     description
25         "Counter that increments every time a Pdelay_Req message is
26         received.";
27     reference
28         "14.12.5 of IEEE Std 802.1AS";
29 }
30 leaf rx-pdelay-resp-count {
31     type yang:counter32;
32     config false;
33     description
34         "Counter that increments every time a Pdelay_Resp message is
35         received.";
36     reference
37         "14.12.6 of IEEE Std 802.1AS";
38 }
39 leaf rx-pdelay-resp-follow-up-count {
40     type yang:counter32;
41     config false;
42     description
43         "Counter that increments every time a Pdelay_Resp_Follow_Up
44         message is received.";
45     reference
46         "14.12.7 of IEEE Std 802.1AS";
47 }
48 leaf rx-announce-count {
49     type yang:counter32;
50     config false;
51     description
52         "Counter that increments every time an Announce message is
53         received.";
54     reference
55         "14.12.8 of IEEE Std 802.1AS";
56 }
57 leaf rx-packet-discard-count {
58     type yang:counter32;
59     config false;

```

```

1      1      description
2      2      "Counter that increments every time a PTP message of the
3      3      respective PTP Instance is discarded.";
4      4      reference
5      5      "14.12.9 of IEEE Std 802.1AS";
6      6      }
7      7      leaf sync-receipt-timeout-count {
8      8      type yang:counter32;
9      9      config false;
10     10     description
11     11     "Counter that increments every time a sync receipt timeout
12     12     occurs.";
13     13     reference
14     14     "14.12.10 of IEEE Std 802.1AS";
15     15     }
16     16     leaf announce-receipt-timeout-count {
17     17     type yang:counter32;
18     18     config false;
19     19     description
20     20     "Counter that increments every time an announce receipt timeout
21     21     occurs.";
22     22     reference
23     23     "14.12.11 of IEEE Std 802.1AS";
24     24     }
25     25     leaf pdelay-allowed-lost-exceeded-count {
26     26     type yang:counter32;
27     27     config false;
28     28     description
29     29     "Counter that increments every time the value of the variable
30     30     lostResponses exceeds the value of the variable
31     31     allowedLostResponses, in the RESET state of the MDPdelayReq
32     32     state machine.";
33     33     reference
34     34     "14.12.12 of IEEE Std 802.1AS";
35     35     }
36     36     leaf tx-sync-count {
37     37     type yang:counter32;
38     38     config false;
39     39     description
40     40     "Counter that increments every time synchronization information
41     41     is transmitted.";
42     42     reference
43     43     "14.12.13 of IEEE Std 802.1AS";
44     44     }
45     45     leaf tx-one-step-sync-count {
46     46     type yang:counter32;
47     47     config false;
48     48     description
49     49     "Counter that increments every time a one-step Sync message is
50     50     transmitted.";
51     51     reference
52     52     "14.12.14 of IEEE Std 802.1AS";
53     53     }
54     54     leaf tx-follow-up-count {
55     55     type yang:counter32;
56     56     config false;
57     57     description
58     58     "Counter that increments every time a Follow_Up message is
59     59     transmitted.";

```

```

1      reference
2      "14.12.15 of IEEE Std 802.1AS";
3  }
4  leaf tx-pdelay-req-count {
5      type yang:counter32;
6      config false;
7      description
8          "Counter that increments every time a Pdelay_Req message is
9          transmitted.";
10     reference
11         "14.12.16 of IEEE Std 802.1AS";
12 }
13 leaf tx-pdelay-resp-count {
14     type yang:counter32;
15     config false;
16     description
17         "Counter that increments every time a Pdelay_Resp message is
18         transmitted.";
19     reference
20         "14.12.17 of IEEE Std 802.1AS";
21 }
22 leaf tx-pdelay-resp-follow-up-count {
23     type yang:counter32;
24     config false;
25     description
26         "Counter that increments every time a Pdelay_Resp_Follow_Up
27         message is transmitted.";
28     reference
29         "14.12.18 of IEEE Std 802.1AS";
30 }
31 leaf tx-announce-count {
32     type yang:counter32;
33     config false;
34     description
35         "Counter that increments every time an Announce message is
36         transmitted.";
37     reference
38         "14.12.19 of IEEE Std 802.1AS";
39 }
40 }
41 }
42
43 augment "/ptp-tt:ptp"
44     + "/ptp-tt:instances"
45     + "/ptp-tt:instance"
46     + "/ptp-tt:ports"
47     + "/ptp-tt:port" {
48     description
49         "Augment to add asymmetry-measurement-mode-ds to IEEE Std 1588 PTP
50         Port.";
51     container asymmetry-measurement-mode-ds {
52         description
53             "Represents the capability to enable/disable the Asymmetry
54             Compensation Measurement Procedure on a PTP Port. This data set
55             is used instead of the CMLDS asymmetry-measurement-mode-ds when
56             only a single PTP Instance is present (i.e., CMLDS is not used).";
57         reference
58             "14.15 of IEEE Std 802.1AS
59             Annex G of IEEE Std 802.1AS";

```

```

1      1      leaf enabled {
2      2          type boolean;
3      3          description
4      4              "For full-duplex IEEE Std 802.3 media, the value is true if an
5      5                  asymmetry measurement is being performed for the link attached
6      6                  to this PTP Port, and false otherwise. For all other media, the
7      7                  value shall be false.";
8      8          reference
9      9              "14.15.2 of IEEE Std 802.1AS";
10     10     }
11     11 }
12     12 }
13     13 augment "/ptp-tt:ptp"
14     14     + "/ptp-tt:common-services"
15     15     + "/ptp-tt:cmlds"
16     16     + "/ptp-tt:ports"
17     17     + "/ptp-tt:port"
18     18     + "/ptp-tt:link-port-ds" {
19     19     description
20     20         "Augment IEEE Std 1588 cmldsLinkPortDS.";
21     21     leaf cmlds-link-port-enabled {
22     22         type boolean;
23     23         config false;
24     24         description
25     25             "Boolean that is true if both delay-mechanism is common-p2p and
26     26                 the value of ptp-port-enabled is true, for at least one PTP Port
27     27                 that uses the CMLDS; otherwise, the value is false.";
28     28         reference
29     29             "11.2.18.1 of IEEE Std 802.1AS
30     30             14.18.3 of IEEE Std 802.1AS";
31     31     }
32     32     leaf is-measuring-delay {
33     33         type boolean;
34     34         config false;
35     35         description
36     36             "This leaf is analogous to is-measuring-delay for a PTP Port, but
37     37                 applicable to this Link Port.";
38     38         reference
39     39             "14.18.4 of IEEE Std 802.1AS";
40     40     }
41     41     leaf as-capable-across-domains {
42     42         type boolean;
43     43         config false;
44     44         description
45     45             "This leaf is true when all PTP Instances (domains) for this Link
46     46                 Port detect proper exchange of Pdelay messages.";
47     47         reference
48     48             "11.2.2 of IEEE Std 802.1AS
49     49             14.18.5 of IEEE Std 802.1AS";
50     50     }
51     51     leaf mean-link-delay-thresh {
52     52         type ptp-tt:time-interval;
53     53         description
54     54             "Propagation time threshold for mean-link-delay, above which a
55     55                 Link Port is not considered capable of participating in the IEEE
56     56                 Std 802.1AS protocol.";
57     57         reference
58     58             "14.18.7 of IEEE Std 802.1AS";
59     59     }

```

```

1      }
2      leaf neg-mean-link-delay-thresh {
3          type ptp-tt:time-interval;
4          description
5              "The negative propagation time threshold for mean-link-delay,
6              below which a Link Port is not considered capable of
7              participating in the IEEE Std 802.1AS protocol.";
8          reference
9              "14.18.8 of IEEE Std 802.1AS";
10     }
11     leaf initial-log-pdelay-req-interval {
12         type int8;
13         description
14             "This leaf is analogous to initial-log-pdelay-req-interval for a
15             PTP Port, but applicable to this Link Port.";
16         reference
17             "14.18.11 of IEEE Std 802.1AS";
18     }
19     leaf current-log-pdelay-req-interval {
20         type int8;
21         config false;
22         description
23             "This leaf is analogous to current-log-pdelay-req-interval for a
24             PTP Port, but applicable to this Link Port.";
25         reference
26             "14.18.12 of IEEE Std 802.1AS";
27     }
28     leaf use-mgt-log-pdelay-req-interval {
29         type boolean;
30         description
31             "This leaf is analogous to use-mgt-log-pdelay-req-interval for a
32             PTP Port, but applicable to this Link Port.";
33         reference
34             "14.18.13 of IEEE Std 802.1AS";
35     }
36     leaf mgt-log-pdelay-req-interval {
37         type int8;
38         description
39             "This leaf is analogous to mgt-log-pdelay-req-interval for a PTP
40             Port, but applicable to this Link Port.";
41         reference
42             "14.18.14 of IEEE Std 802.1AS";
43     }
44     leaf initial-compute-neighbor-rate-ratio {
45         type boolean;
46         description
47             "This leaf is analogous to initial-compute-neighbor-rate-ratio
48             for a PTP Port, but applicable to this Link Port.";
49         reference
50             "14.18.15 of IEEE Std 802.1AS";
51     }
52     leaf current-compute-neighbor-rate-ratio {
53         type boolean;
54         config false;
55         description
56             "This leaf is analogous to current-compute-neighbor-rate-ratio
57             for a PTP Port, but applicable to this Link Port.";
58         reference
59             "14.18.16 of IEEE Std 802.1AS";

```

```

1      1      }
2      2      leaf use-mgt-compute-neighbor-rate-ratio {
3      3          type boolean;
4      4          description
5      5              "This leaf is analogous to use-mgt-compute-neighbor-rate-ratio
6      6                  for a PTP Port, but applicable to this Link Port.";
7      7          reference
8      8              "14.18.17 of IEEE Std 802.1AS";
9      9      }
10     10     leaf mgt-compute-neighbor-rate-ratio {
11     11         type boolean;
12     12         description
13     13             "This leaf is analogous to mgt-compute-neighbor-rate-ratio for a
14     14                 PTP Port, but applicable to this Link Port.";
15     15         reference
16     16             "14.18.18 of IEEE Std 802.1AS";
17     17     }
18     18     leaf initial-compute-mean-link-delay {
19     19         type boolean;
20     20         description
21     21             "This leaf is analogous to initial-compute-mean-link-delay for a
22     22                 PTP Port, but applicable to this Link Port.";
23     23         reference
24     24             "14.18.19 of IEEE Std 802.1AS";
25     25     }
26     26     leaf current-compute-mean-link-delay {
27     27         type boolean;
28     28         config false;
29     29         description
30     30             "This leaf is analogous to current-compute-mean-link-delay for a
31     31                 PTP Port, but applicable to this Link Port.";
32     32         reference
33     33             "14.18.20 of IEEE Std 802.1AS";
34     34     }
35     35     leaf use-mgt-compute-mean-link-delay {
36     36         type boolean;
37     37         description
38     38             "This leaf is analogous to use-mgt-compute-mean-link-delay for a
39     39                 PTP Port, but applicable to this Link Port.";
40     40         reference
41     41             "14.18.21 of IEEE Std 802.1AS";
42     42     }
43     43     leaf mgt-compute-mean-link-delay {
44     44         type boolean;
45     45         description
46     46             "This leaf is analogous to mgt-compute-mean-link-delay for a PTP
47     47                 Port, but applicable to this Link Port.";
48     48         reference
49     49             "14.18.22 of IEEE Std 802.1AS";
50     50     }
51     51     leaf allowed-lost-responses {
52     52         type uint8;
53     53         description
54     54             "This leaf is analogous to allowed-lost-responses for a PTP Port,
55     55                 but applicable to this Link Port.";
56     56         reference
57     57             "14.18.23 of IEEE Std 802.1AS";
58     58     }
59     59     leaf allowed-faults {

```

```

1      type uint8;
2      description
3          "This leaf is analogous to allowed-faults for a PTP Port, but
4          applicable to this Link Port.";
5      reference
6          "14.18.24 of IEEE Std 802.1AS";
7  }
8  leaf-list pdelay-truncated-timestamps {
9      type uinteger48;
10     config false;
11     description
12         "This leaf is analogous to pdelay-truncated-timestamps for a PTP
13         Port, but applicable to this Link Port.";
14     reference
15         "14.18.26 of IEEE Std 802.1AS";
16 }
17 }
18
19 augment "/ptp-tt:ptp"
20     + "/ptp-tt:common-services"
21     + "/ptp-tt:cmllds"
22     + "/ptp-tt:ports"
23     + "/ptp-tt:port" {
24     description
25         "Augment to add port-statistics-ds to IEEE Std 1588 Link Port.";
26     container port-statistics-ds {
27         description
28             "This container is analogous to port-statistics-ds for a PTP
29             Port, but applicable to this Link Port.";
30         reference
31             "14.19 of IEEE Std 802.1AS";
32         leaf rx-pdelay-req-count {
33             type yang:counter32;
34             config false;
35             description
36                 "This leaf is analogous to rx-pdelay-req-count for a PTP Port,
37                 but applicable to this Link Port.";
38             reference
39                 "14.19.2 of IEEE Std 802.1AS";
40         }
41         leaf rx-pdelay-resp-count {
42             type yang:counter32;
43             config false;
44             description
45                 "This leaf is analogous to rx-pdelay-resp-count for a PTP Port,
46                 but applicable to this Link Port.";
47             reference
48                 "14.19.3 of IEEE Std 802.1AS";
49         }
50         leaf rx-pdelay-resp-follow-up-count {
51             type yang:counter32;
52             config false;
53             description
54                 "This leaf is analogous to rx-pdelay-resp-follow-up-count for a
55                 PTP Port, but applicable to this Link Port.";
56             reference
57                 "14.19.4 of IEEE Std 802.1AS";
58         }
59         leaf rx-packet-discard-count {

```

```

1      1      type yang:counter32;
2      2      config false;
3      3      description
4      4          "This leaf is analogous to rx-packet-discard-count for a PTP
5      5          Port, but applicable to this Link Port.";
6      6      reference
7      7          "14.19.5 of IEEE Std 802.1AS";
8      8      }
9      9      leaf pdelay-allowed-lost-exceeded-count {
10     10         type yang:counter32;
11     11         config false;
12     12         description
13     13             "This leaf is analogous to pdelay-allowed-lost-exceeded-count
14     14             for a PTP Port, but applicable to this Link Port.";
15     15         reference
16     16             "14.19.6 of IEEE Std 802.1AS";
17     17     }
18     18     leaf tx-pdelay-req-count {
19     19         type yang:counter32;
20     20         config false;
21     21         description
22     22             "This leaf is analogous to tx-pdelay-req-count for a PTP Port,
23     23             but applicable to this Link Port.";
24     24         reference
25     25             "14.19.7 of IEEE Std 802.1AS";
26     26     }
27     27     leaf tx-pdelay-resp-count {
28     28         type yang:counter32;
29     29         config false;
30     30         description
31     31             "This leaf is analogous to tx-pdelay-resp-count for a PTP Port,
32     32             but applicable to this Link Port.";
33     33         reference
34     34             "14.19.8 of IEEE Std 802.1AS";
35     35     }
36     36     leaf tx-pdelay-resp-follow-up-count {
37     37         type yang:counter32;
38     38         config false;
39     39         description
40     40             "This leaf is analogous to tx-pdelay-resp-follow-up-count for a
41     41             PTP Port, but applicable to this Link Port.";
42     42         reference
43     43             "14.19.10 of IEEE Std 802.1AS";
44     44     }
45     45 }
46     46 }
47
48     48 augment "/ptp-tt:ptp"
49     49     + "/ptp-tt:common-services"
50     50     + "/ptp-tt:cmls"
51     51     + "/ptp-tt:ports"
52     52     + "/ptp-tt:port" {
53     53     description
54     54         "Augment to add asymmetry-measurement-mode-ds to IEEE Std 1588 Link
55     55         Port.";
56     56     container asymmetry-measurement-mode-ds {
57     57         description
58     58             "This container is analogous to asymmetry-measurement-mode-ds for
59     59             a PTP Port, but applicable to this Link Port.";

```

```
1      reference
2      "14.20 of IEEE Std 802.1AS";
3      leaf enabled {
4          type boolean;
5          description
6              "This leaf is analogous to
7              asymmetry-measurement-mode-ds.enabled for a PTP Port, but
8              applicable to this Link Port.";
9          reference
10         "14.20.2 of IEEE Std 802.1AS";
11     }
12 }
13 }
14 }
15
16
```

1 *Insert the following new Clause 19:*

2 **19. Media-dependent layer specification for IEEE 802.3 Clause 4 Media** 3 **Access Control (MAC) operating in half-duplex**

4 **19.1 Overview**

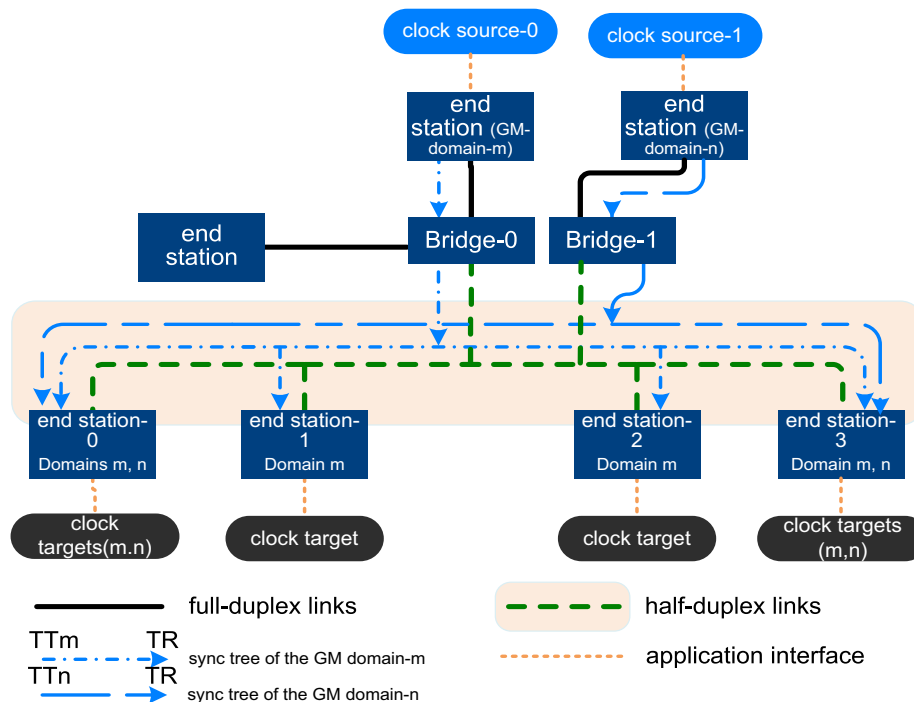
5 **19.1.1 General**

6 This clause specifies the media-dependent layer that provides synchronized time across links using the IEEE
7 Std 802.3 (Ethernet) Clause 4 MAC operating in half-duplex mode

8 **19.1.1.1 Half-duplex Ethernet (HDE) characteristics**

9 The Ethernet IEEE 802.3 Clause 4 MAC can operate in either full-duplex or half-duplex mode. When this
10 MAC is operating in full-duplex, its media-dependent specification for gPTP is covered in Clause 11 of this
11 standard. The present clause is used when the 802.3 Clause 4 MAC is operating in half-duplex. This mode
12 necessitates additional managed object settings and frame processing due to the effects of the shared media
13 this mode supports, which are described in this clause.

14 Figure 19-1 shows an example of time-aware network using HDE.



Note 1: The “bridge” in this figure is an example of a time-aware system that contains a PTP Relay Instance, and the end stations with clock targets are examples of time-aware systems that contain PTP End Instances.
Note 2: GM denotes Grandmaster PTP Instance

Figure 19-1— Time-aware network example where the four end stations near the bottom of the figure are connected to the bridge via an HDE link

1 When using HDE, the use of “neighbor” refers to a device or to devices (end station or bridge) that is/are the
2 intended recipient(s) of a transmitted PTP message. The intended recipient(s) of a PTP message from a
3 gPTP TT port is/are one or more gPTP TR port(s). The intended recipient of a PTP message from a gPTP
4 TR port is the gPTP TT port, regardless of whether or not other gPTP TT ports received that message (e.g.,
5 if all ports are connected via a shared medium). In the example in Figure 19-1, PTP messages are exchanged
6 between a gPTP TT port of the Bridge and the four gPTP TR ports of the four end stations via a shared
7 medium.

8 NOTE—For example, this standard supports Physical Layer Collision Avoidance (PLCA) (10BASE-T1S) as specified
9 in Clause 147 of IEEE Std 802.3.

10 **19.1.1.2 Overview of the major differences and restrictions of using HDE**

11 The present clause (Clause 19) uses Clause 11 in its entirety with the following major differences and
12 restrictions (complete details follow starting in 19.1.2):

- 13 a) The peer delay initiator is restricted to timeReceivers only (see 19.1.2).
- 14 b) One-step time transport is not supported (see 19.1.3 and 19.2.16)
- 15 c) CMLDS is not supported (see 19.2.17)
- 16 d) External port configuration mode is the only mode supported (see 19.8)
- 17 e) The use of Signaling messages is not specified (see 19.8)
- 18 f) Hot Standby is not supported (see 19.8)

19 **19.1.2 Propagation delay measurement over links**

20 The measurement of propagation delay on an HDE using the peer-to-peer delay mechanism is illustrated in
21 Figure 11-1 and is described in 11.1.2, with the following differences:

- 22 a) The peer delay initiator is restricted to each timeReceiver port
- 23 b) The timeTransmitter port does not initiate the peer-to-peer delay mechanism.

24 Therefore, pdelayReqSendDisabled and pdelayRespSendDisabled are set as follows:

- 25 c) pdelayReqSendDisabled shall be set to TRUE for a timeTransmitter port or a PassivePort.
- 26 d) pdelayRespSendDisabled shall be set to TRUE for a timeReceiver port or a PassivePort.

27 **19.1.3 Transport of time-synchronization information**

28 HDE shall use two-step time transport as specified in clause 11.1.3. The variables oneStepReceive (see
29 19.2.13.9), oneStepTransmit (see 19.2.13.10), and oneStepTxOper (see 19.2.13.11) shall be set to FALSE for
30 HDE

31 NOTE—The transport of time-synchronization information by a PTP Instance, using Sync and Follow_Up messages, is
32 illustrated in Figure 11-2.

33 **19.1.4 Model of operation**

34 The model for a PTP Instance of a time-aware system with full-duplex point-to-point links is shown in
35 Figure 11-3. This clause reuses Figure 11-3 (as its structure is unchanged for this clause), where all
36 references to clause 11 in Figure 11-3 are to be replaced by references to clause 19 (this clause). The
37 presence of one HDE MD entity per PTP Port is assumed.

1 A general, media-independent description of the generation of timestamps is given in 8.4.3. A more specific
2 description for PTP event messages is given in 11.3.2.1. A PTP event message is timestamped relative to the
3 LocalClock entity when the message timestamp point (see 3.17) crosses the timestamp measurement plane
4 (see 3.33). The timestamp is corrected for any ingressLatency or egressLatency (see 8.4.3) to produce a
5 timestamp relative to the reference plane (see 3.26). The corrected timestamp value is provided to the MD
6 entity.

7 The MD entity behavior and detailed state machines specific to full-duplex point-to-point links, which are
8 described in 11.2, are reused for HDE subject to the conditions defined in 19.2.

9 **19.2 State machines for HDE**

10 **19.2.1 General**

11 The state machines for HDE are described in 11.2.1.

12 **19.2.2 Determination of asCapable and asCapableAcrossDomains**

13 Determination of asCapableAcrossDomains shall be as specified in 11.2.13.12.

14 NOTE—For full-duplex point-to-point links, asCapableAcrossDomains is one global variable for all domains per
15 linkport. For HDE, asCapableAcrossDomains is a global variable per PTP Instance (i.e., per domain), the name
16 asCapableAcrossDomains has been kept for backwards compatibility with existing implementations.

17 **19.2.3 Use of MAC Control PAUSE operation**

18 This is not applicable when the IEEE 802.3 Clause 4 MAC is in half-duplex mode (see Clause 1.4.458 in
19 Annex 31B of IEEE Std 802.3).

20 **19.2.4 Use of priority-based flow control**

21 This is not applicable when the IEEE 802.3 Clause 4 MAC is in half-duplex mode (see Clause 1.4.489 of
22 IEEE Std 802.3).

23 **19.2.5 Use of link aggregation**

24 This is not applicable when the IEEE 802.3 Clause 4 MAC is in half-duplex mode (see Introduction in IEEE
25 Std 802.1AX-2020).

26 **19.2.6 Service interface primitives and data structures communicated between state 27 machines**

28 The following two subclauses describe the service primitives and data structures communicated between the
29 time-synchronization state machines of the MD entity. First the service primitives are described, followed by
30 the data structures.

31 **19.2.7 DL-UNITDATA.request**

32 This service primitive is specified in 11.2.7.

33 **19.2.8 DL-UNITDATA.indication**

34 This service primitive is specified in 11.2.8.

1 **19.2.9 MDTimestampReceive**

2 This data structure shall be as specified in in 11.2.9. ■

3 **19.2.10 MDSyncReceive**

4 This data structure shall be as specified in 10.2.2.2. ■

5 **19.2.11 MDSyncSend**

6 This structure shall be as specified in 10.2.2.1.

7 **19.2.12 Overview of MD entity global variables**

8 The overview of MD entity global variables is given in 11.2.12.

9 **19.2.13 MD entity global variables**

10 **19.2.13.1 currentLogPdelayReqInterval:** This variable shall be as specified in 11.2.13.1.

11 **19.2.13.2 initialLogPdelayReqInterval:** This variable shall be as specified in 11.2.13.2.

12 **19.2.13.3 pdelayReqInterval:** This variable shall be as specified in 11.2.13.3.

13 **19.2.13.4 allowedLostResponses:** This variable shall be as specified in 11.2.13.4.

14 **19.2.13.5 allowedFaults:** This variable shall be as specified in 11.2.13.5.

15 **19.2.13.6 isMeasuringDelay:** This variable shall be as specified in 11.2.13.6.

16 **19.2.13.7 meanLinkDelayThresh:** This variable shall be as specified in 11.2.13.7.

17 **19.2.13.8 syncSequenceId:** This variable shall be as specified in 11.2.13.8.

18 **19.2.13.9 oneStepReceive:** This variable shall be as specified in 11.2.13.9. ■

19 **19.2.13.10 oneStepTransmit:** This variable shall be as specified in 11.2.13.10.

20 **19.2.13.11 oneStepTxOper:** This variable shall be as specified in 11.2.13.11. ■

21 **19.2.13.12 asCapableAcrossDomains:** This variable shall be as specified in 11.2.13.12.

22 **19.2.13.13 nrrPdelay:** This variable shall be as specified in 11.2.13.13.

23 **19.2.13.14 nrrSync:** This variable shall be as specified in 11.2.13.14.

24 **19.2.13.15 nrrCompMethod:** This variable shall be as specified in 11.2.13.15.

25 **19.2.14 MDSyncReceiveSM state machine**

26 The MDSyncReceiveSM state machine shall be as specified in 11.2.14.

1 **19.2.15 MDSyncSendSM state machine**

2 The MDSyncSendSM state machine shall be as specified in 11.2.15.

3 **19.2.16 OneStepTxOperSetting state machine**

4 This state machine is not used for HDE.

5 **19.2.17 Common Mean Link Delay Service (CMLDS)**

6 The Common Mean Link Delay Service (CMLDS) shall not be used for HDE.

7 HDE uses the transport-specific peer-to-peer delay mechanism for each domain. Therefore, if the time-
8 aware system implements multiple domains on an HDE link, each domain uses the transport-specific peer-
9 to-peer delay mechanism on that link regardless of its domain number.

10 If multiple TimeTransmitter ports are present on an HDE link, they are in different gPTP domains. CMLDS
11 cannot be used because, in general, the TimeTransmitter ports can be on physical ports of different time-
12 aware systems (i.e., different bridges). In this case, both meanLinkDelay and neighborRateRatio between a
13 PTP End Instance in one of the domains and the TimeTransmitter it is communicating with can be different
14 from meanLinkDelay and neighborRateRatio between a PTP End Instance in another domain on the same
15 end station and the TimeTransmitter that PTP End Instance is communicating with. For instance, in Figure
16 19-1 the meanLinkDelay and neighborRateRatio between the PTP End Instance over domain mand over
17 domain-n at end station-0 are computed using timestamps from two distinct time aware systems, GM-
18 domain-m and GM-domain-n, respectively; and hence does not use CMLDS.

19 **19.2.18 Common Mean Link Delay Service (CMLDS) global variables**

20 The Common Mean Link Delay Service (CMLDS) global variables are not used for HDE.

21 **19.2.19 MDPdelayReq state machine**

22 The MDPdelayReq state machine shall be as specified in 11.2.19.

23 The variable pdelayReqSendDisabled is set per 19.1.2.

24 **19.2.20 MDPdelayResp state machine**

25 The MDPdelayResp state machine shall be as specified in 11.2.20.

26 The variable pdelayRespSendDisabled is set per 19.1.2.

27 **19.2.21 LinkDelayIntervalSetting state machine**

28 This state machine is not used for HDE.

29 **19.3 Message attributes**

30 Message attributes shall be as specified in 11.3.

1 19.4 Message formats

2 Message formats shall be as specified in 11.4, except for 11.4.2.4, where the domainNumber for
3 Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be the domain number of the HDE
4 gPTP domain used by the transport-specific peer delay mechanism. The domainNumber for all other PTP
5 messages is as specified in 10.6.2.2.6.

6 19.5 Protocol timing characterization

7 19.5.1 General

8 This subclause specifies timing attributes for the media-dependent sublayer specified in this clause.

9 19.5.2 Message transmission intervals

10 19.5.2.1 General interval specification

11 The mean time interval between successive Pdelay_Req messages is represented as the logarithm to the
12 base 2 of this time interval measured in seconds. The value of this logarithmic attribute shall be as specified
13 in 19.5.2.2.

14 The mean time interval between successive Sync messages shall be as specified in 10.7.2.1, 10.7.2.3, and
15 19.5.2.3.

16 19.5.2.2 Pdelay_Req message transmission interval

17 Pdelay_Req message transmission interval is specified in 11.5.2.2. The variable
18 useMgtSettableLogPdelayReqInterval shall be set to TRUE.

19 19.5.2.3 Sync message transmission interval default value

20 The Sync message transmission interval default value shall be as specified in 11.5.2.3.

21 In the case of non-deterministic behavior (see IEEE Std 802.3) in an HDE link, the nondeterministic
22 behavior needs to be considered in determining whether the requirements of 10.7.2.3 for the Sync message
23 transmission interval are met.

24 19.5.3 allowedLostResponses

25 The variable allowedLostResponses shall be as specified in 11.5.3.

26 19.5.4 allowedFaults

27 The variable allowedFaults shall be as specified in 11.5.4.

28 19.6 Control of computation of neighborRateRatio

29 The control of computation of neighborRateRatio shall be as specified in 11.6.

30 19.7 Control of computation of meanLinkDelay

31 The control of computation of meanLinkDelay shall be as specified in 11.7.

1 19.8 HDE settings and configuration

2 This subclause provides settings and configurations that are specific for HDE.

3 The per PTP Instance global variable `externalPortConfigurationEnabled` shall be set to TRUE for HDE.

4 Both `GtpCapableTransmit` and `GtpCapableReceive` state machines shall be disabled for HDE, and
5 therefore `gtpCapableStateMachinesEnabled` shall be set to FALSE (see 10.4.1). According to 10.4.1, if the
6 managed object `gtpCapableStateMachinesEnabled` is FALSE, the global variable `neighborGtpCapable` for
7 the port (see 10.2.5.16) is set to TRUE.

8 Hot Standby (see Clause 18) is not supported on HDE.

9 The `SyncIntervalSetting` and the `AnnounceIntervalSetting` state machines are not used for HDE. The
10 variables `useMgtSettableLogSyncInterval` and `useMgtSettableLogAnnounceInterval` shall be set to TRUE.

11 The `GtpCapableIntervalSetting` state machine is not used for HDE. The variable
12 `useMgtSettableLogGtpCapableMessageInterval` shall be set to TRUE.

Annex A

(normative)

Protocol Implementation Conformance Statement (PICS) proforma³

Add a row at the end of Table A.5 as follows:

A.5 Major Capabilities

Item	Feature	Status	References	Support
MDHDE	Does the PTP Instance support media-dependent HDE link functionality on one or more PTP Ports?	O:1	5.9, 11, 19, A.6, A.22	Yes [] No []

Change A.6 as follows:

A.6 Media access control methods

Item	Feature	Status	References	Support
MAC-IEEE-802.3	Which MAC methods are implemented in conformance with the relevant MAC standards?	O:2	11.1 19.1	Yes [] No [] Yes [] No []
MAC-IEEE-802.11		O:2	12.1	Yes [] No []
MAC-1	Has a PICS been completed for each of the MAC methods implemented as required by the relevant MAC Standards?	M		Yes []
MAC-2	Do all the MAC methods implemented support the MAC Timing aware Service as specified?	M	Clause 11 Clause 12 Clause 13 Clause 19	Yes []

³ Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

Change A.21 as follows:**A.21 External port configuration**

Item	Feature	Status	References	Support
	If item EXT is not supported, mark N/A.			NA []
EXT-1	Does the PTP Instance support the specifications for externalPortConfigurationEnabled value of true?	EXT:M	10.3.1 19.8	Yes []
EXT-2	Does the PTP Instance support the PortAnnounceInformationExt state machine?	EXT:M	10.3.14	Yes []
EXT-3	Does the PTP Instance support the PortStateSettingExt state machine?	EXT:M	10.3.15	Yes []

<<Editor's note: Table A.22 is based on Table A.13 for full-duplex point-to-point link. Please, check.>

Insert Table A.22 as follows:**A.22 Media-dependent, HDE link**

Item	Feature	Status	References	Support
MDHDE-1	Does this PTP Port implement the functionality of the MDSyncReceiveSM state machine in compliance with the requirements of 19.2.14, 11.2.14 and Figure 11-6?	MDHDE:M	19.2.14, 11.2.14	Yes []
MDHDE-2	Does this PTP Port implement the functionality of the MDSyncSendSM state machine in compliance with the requirements of 19.2.15, 11.2.15 and Figure 11-7?	MIMSTR and MDHDE:M	19.2.15, 11.2.15	Yes []
MDHDE-3	Does this port implement the functionality of the MDPdelayRequest state machine in compliance with the requirements of 19.1.2, 19.2.19, 11.2.19 and Figure 11-9?	MDHDE:M	19.1.2, 19.2.19, 11.2.19	Yes []
MDHDE-4	Does this port implement the functionality of the MDPdelayResponse state machine in compliance with the requirements of 19.1.2, 19.2.20, 11.2.20 and Figure 11-10?	MDHDE:M,	19.2.20 19.1.2 11.2.20	Yes []
MDHDE-5	Does this PTP Port timestamp Sync messages on ingress with respect to the LocalClock in compliance with 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes []
MDHDE-6	Does this PTP Port timestamp Sync messages on egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MIMSTR and MDHDE:M	19.3 11.3.2.1	Yes []
MDHDE-7	Does this port timestamp Pdelay_Req messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes []

A.22 Media-dependent, HDE link (continued)

Item	Feature	Status	References	Support
MDHDE-8	Does this port timestamp Pdelay_Resp messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes []
MDHDE-9	Are all IEEE 802.1AS messages on this port sent without a Q-tag in compliance with the requirements of 19.3, 11.3.3?	MDHDE:M	19.3, 11.3.3	Yes []
MDHDE-10	Do all media-dependent messages transmitted on this port use a destination MAC address taken from Table 11-3 in compliance with the requirements of 19.3 and 11.3.4 [01-80-C2-00-00-0E]?	MDHDE:M	19.3, 11.3.4	Yes []
MDHDE-11	Do all media-dependent messages transmitted on this port use a source MAC address that is assigned to that port in compliance with the requirements of 19.3 and 11.3.4?	MDHDE:M	19.3, 11.3.4	Yes []
MDHDE-12	Do all media-dependent message transmitted on this port use an EtherType specified in Table 11-4 [88-F7]?	MDHDE:M	19.3, 11.3.5	Yes []
MDHDE-13	Does the header of all the media-dependent messages on this port comply with the requirements of 19.4, 11.4.2 and Table 10-7?	MDHDE:M	19.4, 11.4.2	Yes [] N/A []
MDHDE-14	Does the body of Sync messages sent on this PTP Port comply with the requirements of 19.4, 11.4.3, Table 11-8, and Table 11-9?	MDHDE:M	19.4, 11.4.3	Yes []
MDHDE-15	Does the body of Follow_Up messages sent on this PTP Port comply with the requirements of 19.4, 11.4.4, 6.4.3.3 (lastGmPhaseChange), and Table 11-10?	MDHDE:M	19.4, 11.4.4, 6.4.3.3	Yes []
MDHDE-16	Does the body of Pdelay_Req messages sent on this port comply with the requirements of 19.4, 11.4.5 and Table 11-12?	MDHDE:M	19.4, 11.4.5	Yes []
MDHDE-17	Does the body of Pdelay_Resp messages sent on this port comply with the requirements of 19.4, 11.4.6 and Table 11-13?	MDHDE:M	19.4, 11.4.6	Yes []
MDHDE-18	Does the body of Pdelay_Resp_Follow_Up messages sent on this port comply with the requirements of 19.4, 11.4.7 and Table 11-14?	MDHDE:M	19.4, 11.4.7	Yes []
MDHDE-19	Are all reserved fields in media-dependent messages sent on this port set to 0 in compliance with the requirements of 19.4, 11.4.1?	MDHDE:M	19.4, 11.4.1	Yes []
MDHDE-20	Do the Sync message sequence numbers comply with the requirements of 19.3, 11.3.8?	MIMSTR and MDHDE:M	19.3, 11.3.8	Yes [] N/A []

A.22 Media-dependent, HDE link (continued)

Item	Feature	Status	References	Support
MDHDE-21	Do the Pdelay_Req message sequence numbers comply with the requirements of 19.3 and 11.3.8?	MDHDE:M	19.3, 11.3.8	Yes []
MDHDE-22	Does the Pdelay mean request transmission interval comply with the requirements of 19.5.2.2 and 11.5.2.2?	MDHDE:M	19.5.2.2, 11.5.2.2	Yes []
MDHDE-23	Does the Sync mean transmission interval comply with the requirements of 19.5.2.3 and 11.5.2.3?	MDHDE:M	19.5.2.3, 11.5.2.3	Yes []
MDHDE-24	Does the HDE media-dependent layer set the asCapable global variable in the media-independent PortSync entity in compliance with the requirements of 19.2.2 and 11.2.2?	MDHDE:M	19.2.2, 11.2.2	Yes []
MDHDE-25	Does the PTP Instance consider the PTP Port or Link Port, respectively, to not be exchanging Pdelay messages when a valid response is not received in compliance with the requirements of 19.5.3 and 11.5.3?	MDHDE:M	19.5.3, 11.5.3	Yes []
MDHDE-26	Does the PTP Instance ignore TLVs, of PTP messages, that it cannot parse and attempt to parse the next TLV, in compliance with the requirements of 19.4 and 11.4.1?	MDHDE:M	19.4, 11.4.1	Yes []
MDHDE-27	Does the time-aware system initialize meanLinkDelayThresh as specified in 19.2.2 and 11.2.2?	MDHDE:M	19.2.2, 11.2.2	Yes []
MDHDE-28	Does this port support propagation delay averaging?	MDHDE:O	19.2.19, 11.2.19.3.4	Yes [] No []
MDHDE-29	Does this port support two-step capability on receive?	MDHDE:M	19.2.14, 11.2.14, item d) of 5.5	Yes []
MDHDE-30	Does this port support two-step capability on transmit?	MDHDE:M	19.2.15, 11.2.15, item e) of 5.5	Yes []
MDHDE-31	Is the transport-specific peer-to-peer delay mechanism supported?	MDHDE:M	19.2.17	Yes []

Annex F

(informative)

PTP profile included in this standard

F.1 General

Change the paragraph in F.1 as follows:

The specification in this standard of synchronized time transport over a full-duplex point-to-point link includes a PTP profile. The information contained in a PTP profile is described in 20.3 of IEEE Std 1588-2019. This annex summarizes the PTP profile for transport of timing over full-duplex point-to-point links. This PTP profile is also used [in the transport of timing over HDE \(see Clause 19\)](#), and in the transport of timing over CSN when a CSN clock reference is not present (see Clause 16). This PTP profile is not used in the transport of timing over IEEE 802.11 links and IEEE 802.3 EPON links; both these transports use native timing mechanisms to assist in the synchronized time transport.

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